Sinais e Sistemas para Engenharia

MÓDULO 5 - ANÁLISE NO TEMPO DE SISTEMAS CONTÍNUOS

VERSÃO OCTAVE

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No módulo anterior, vimos que um sinal discreto x(n) qualquer pode ser expresso como um soma de impulsos com amplitude variável, deslocados no tempo:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

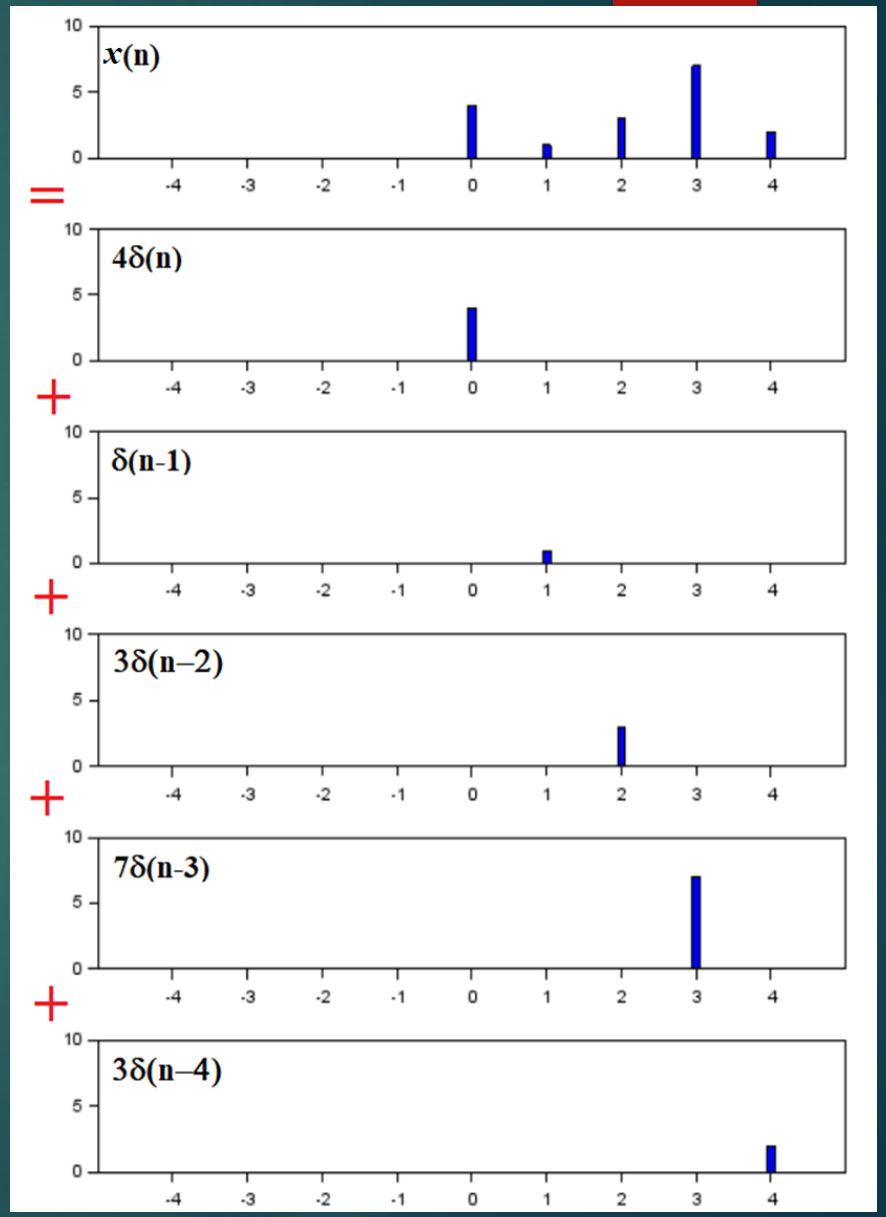
Da mesma maneira, um sinal contínuo $x(t)$ qualquer pode ser expresso como um soma infinita (integral) de impulsos contínuos com amplitude variável, deslocados no tempo

$$x(t) = \int_{-\infty}^{+\infty} x(\tau)\delta(t-\tau)d\tau$$

Exemplo:

$x(n) = [0, 0, 0, 0, 4, 1, 3, 7, 2]$ (-4)

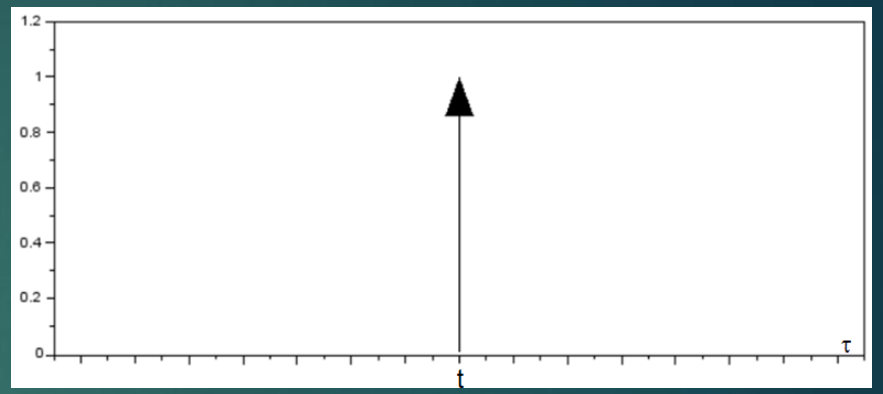
$x(n) = 4\delta(n)+\delta(n-1)+3\delta(n-2)+7\delta(n-3)+2\delta(n-4)$



Prova que um sinal contínuo pode ser expresso pela soma de impulsos deslocados

$$x(t) = \int_{-\infty}^{+\infty} x(\tau)\delta(t-\tau)d\tau$$

O impulso contínuo $\delta(\tau)$ deslocado para tempo $\tau = t$ será:



$$\delta(t-\tau) = \begin{cases} 0; & \tau \neq t \\ \infty; & \tau = t \end{cases}$$

Pela definição, $\int_{-\infty}^{+\infty} \delta(t-\tau)dt = \int_{\tau=t^-}^{\tau=t^+} \delta(t-\tau)dt = 1$

Como o impulso só tem valor não-nulo no intervalo $[t^-, t^+]$, podemos mudar os limites de integração

$$x(t) = \int_{-\infty}^{+\infty} x(\tau)\delta(t-\tau)d\tau = \int_{\tau=t^-}^{\tau=t^+} x(\tau)\delta(t-\tau)d\tau$$

Neste intervalo de integração $[t^-, t^+]$, $t = \tau$, logo $x(\tau)$ pode ser substituído por x(t). Como a integral é em $d\tau$, podemos retirar x(t) da integral

$$x(t) = \int_{\tau=t^-}^{\tau=t^+} x(t)\delta(t-\tau)d\tau = x(t)\int_{\tau=t^-}^{\tau=t^+} \delta(t-\tau)d\tau$$

A integral em amarelo é por definição (definição do impulso contínuo) igual a 1, de modo que:

$x(t) = x(t)$ CQD

Convolução Contínua - Definição

Suponha um sistema contínuo linear invariante no tempo, com função de transferência H(s):

$$x(t) \longrightarrow \boxed{\boldsymbol{H(s)}} \longrightarrow y(t)$$

Se na entrada colocarmos o impulso contínuo δ(t), então a saída será h(t), isto é, a resposta ao impulso do sistema H(s).

$$\delta(t) \longrightarrow \boxed{\boldsymbol{H(s)}} \longrightarrow h(t)$$

Como uma entrada genérica $x(t)$ pode ser decomposta em uma soma infinita (integral) de impulsos deslocado e escalonados, a resposta $y(t)$ do sistema poderá ser calculada pela combinação linear (escalonada e deslocada no tempo) da resposta ao impulso $h(t)$ do sistema contínuo. A esta integral, damos o nome de Convolução. Convolução da entrada pela resposta ao impulso do sistema.

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau \xRightarrow{H(s)} y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

$$\boxed{y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau}$$

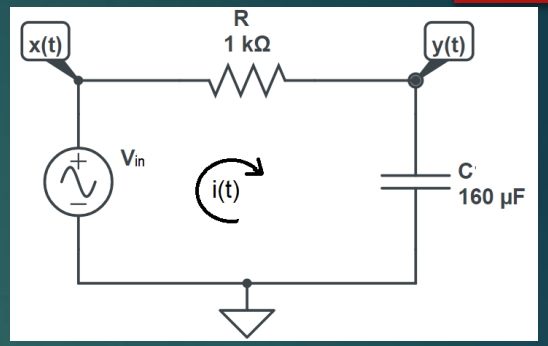
Convolução contínua

$h(t)$ - *Resposta ao impulso do sistema*
$x(t)$ - *Uma entrada qualquer*
$y(t)$ - *Resposta do sistema à entrada x(t)*

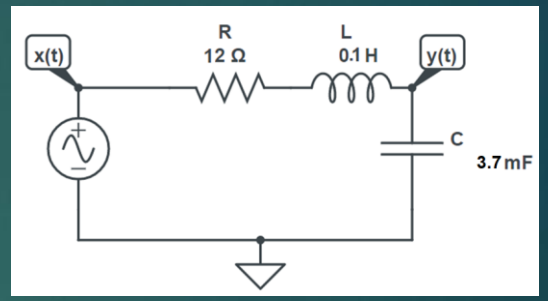
Se conhecermos a resposta ao impulso de um Sistema, podemos obter a sua resposta a qualquer outra entrada através da convolução. Como os sistemas físicos são normalmente descritos por uma equações diferenciais, podemos obter a resposta ao impulso resolvendo a equação diferencial.

Para utilizarmos a convolução contínua e calcularmos a resposta de sistemas contínuos a diversas entradas, analisaremos 3 exemplos:

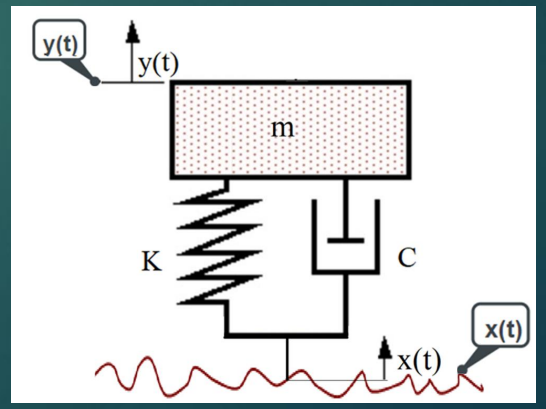
- A) Sistemas de 1ª ordem: Circuito Eletrônico RC



- B) Sistemas de 2ª ordem: : Circuito Eletrônico RLC

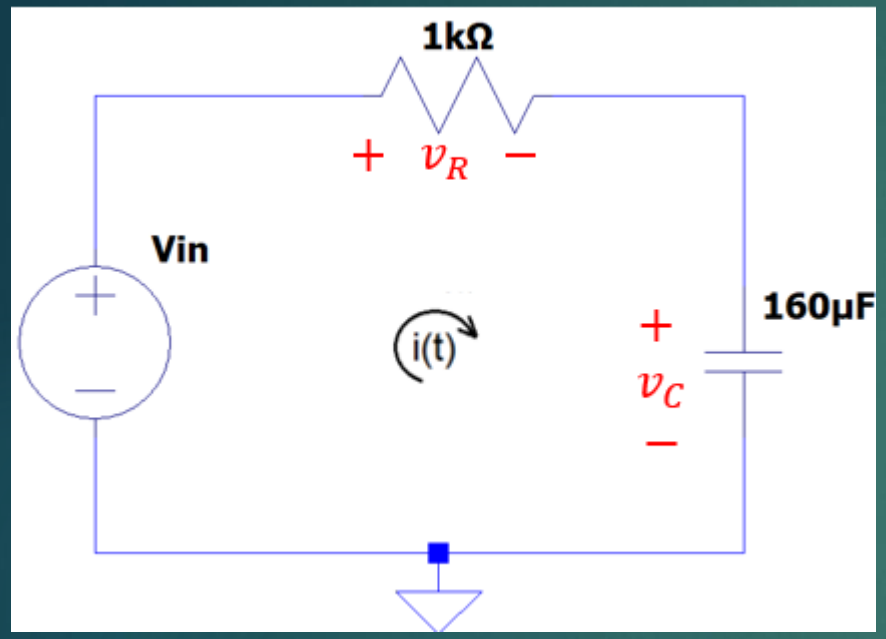


- C) Uma Suspensão Automotiva CKM



A) Resposta de Sistemas de 1ª Ordem a sinais contínuos

Como exemplo, seja o sistema eletrônico RC mostrado abaixo no qual se quer encontrar a tensão no capacitor $v_c(t)$ (saída do do sistema) , resultante da tensão na fonte $v_{in}(t)$ (entrada do sistema)



$R = 1\ k\Omega$
$C = 160\ \mu F$
Condições iniciais $v_C(0^-) = 0$

$Malha \quad -v_{in}(t) + v_R(t) + v_C(t) = 0$

$v_R(t) = R\ i(t)$

$$v_C(t) = \frac{1}{C}\int_o^t i(\tau)d\tau \Rightarrow\ i(t) = C\ v_c'(t)$$

Logo $v_R(t) = R\ i(t) = R\ (Cv_c'(t))$

Então :

$-v_{in}(t) + R\ (Cv_c'(t))\ + v_C(t) = 0$

$$v_c'(t) + \frac{1}{\mathrm{RC}}v_c(t) = \frac{1}{RC}v_{in}(t)$$

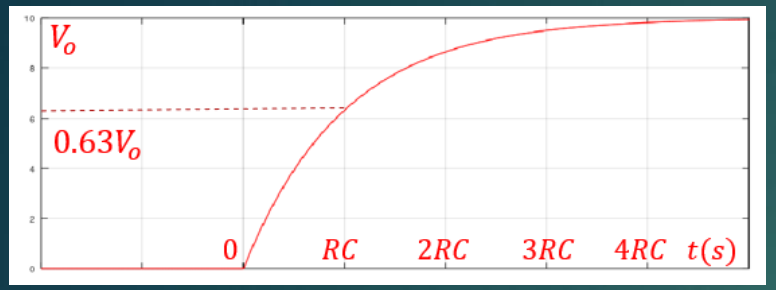
Com : $R = 1\ k\Omega \quad eC = 160\ \mu F$

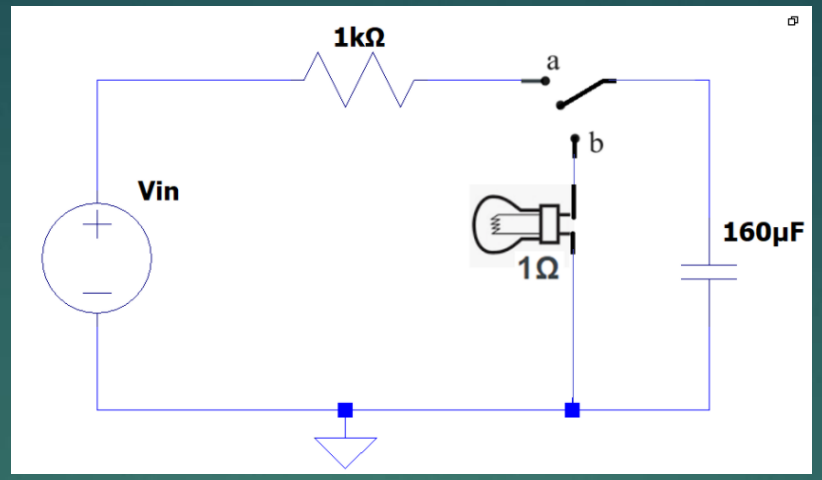
$$v_c'(t) + 6.25v_C(t) = 6.25v_{in}\ (t)$$

$x(t) = v_{in}(t)$
$y(t) = v_c(t)$

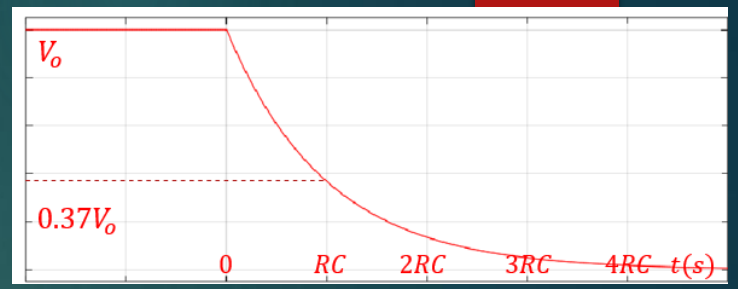
$$y'(t) + 6.25y(t) = 6.25x(t)$$

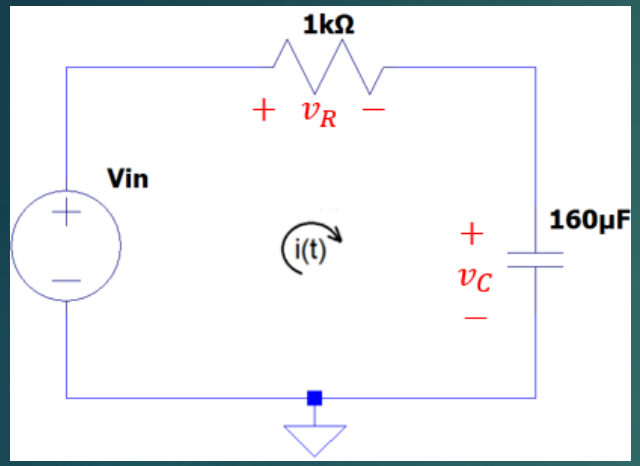
Chave em 'a' - Carregamento



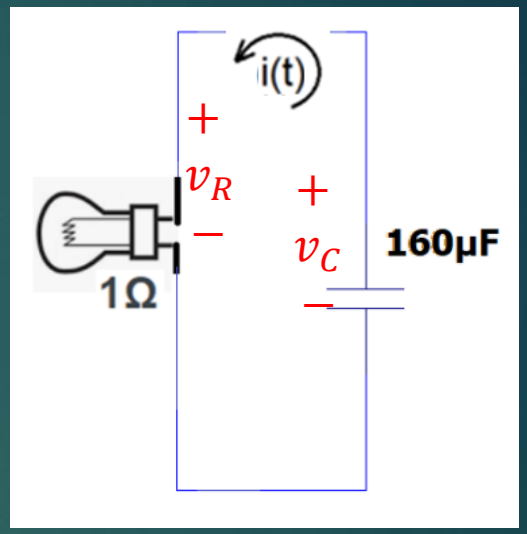


Chave em 'b' - Flash





Este sistema pode representar o carregamento de um Flash de máquina fotográfica



$$v_c'(t) + \frac{1}{\mathrm{RC}} v_c(t) = \frac{1}{RC} v_{in}(t)$$

Como : $RC = 0.16\ s$

$\mathrm{V_{in}} = 300\ V$

$R = 1000\Omega; \quad C = 160\ \mu F$

Condições iniciais $v_C(0^-) = 0$

$$v_c'(t) + \frac{1}{\mathrm{RC}} v_c(t) = 0$$

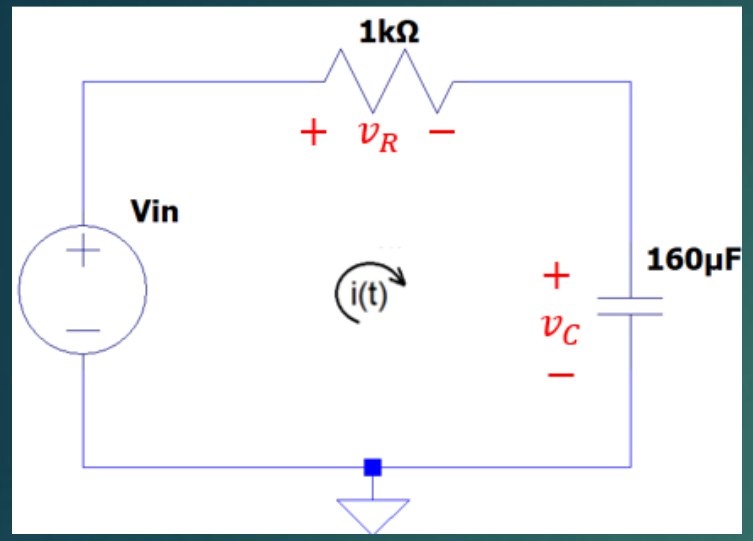
Como : $RC = 0.16ms = 0.00016\ s$

$R = 1\ \Omega$

$C = 160\ \mu F$

Condições iniciais $v_C(0^-) = 300V$

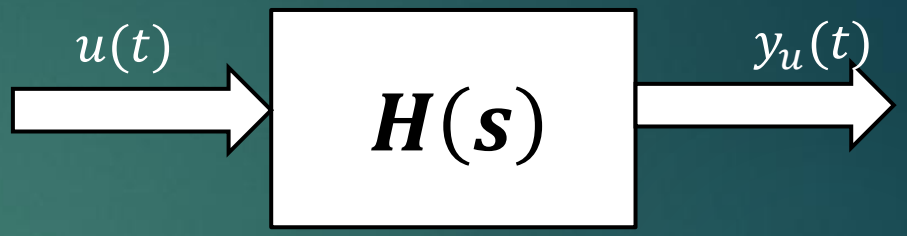
Calcular através da Equação Diferencial a resposta deste sistema para as entradas *Degrau e Impulso*



$$y'(t) + 6.25y(t) = 6.25x(t)$$

$$x(t) \rightarrow \boldsymbol{X}(s)$$
$$x'(t) \rightarrow s\boldsymbol{X}(s)$$
$$y(t) \rightarrow \boldsymbol{Y}(s)$$
$$y'(t) \rightarrow sY(s)$$
$$\boldsymbol{H}(s) = \frac{\boldsymbol{Y}(s)}{\boldsymbol{X}(s)}$$

$$y(0^-) = 0$$



$$y'(t) + 6.25y(t) = 6.25x(t)$$
$$sY(s) + 6.25Y(s) = 6.25X(s)$$
$$H(s) = \frac{Y(s)}{X(s)} = \frac{6.25}{s + 6.25}$$

$$H(s) = \frac{6.25}{s + 6.25}$$

- Calcular primeiro a resposta ao degrau $y_u(t)$

Fazemos $x(t)$=1 com condições iniciais nulas:

- $y_u(0^-) = 0$
- Resposta ao degrau $y(t)$ será a soma da resposta homogênea mais a resposta forçada
- $y_u(t) = y_h(t) + y_f(t)$
- A Reposta ao Impulso $\mathrm{h(t)} = y_o{}'(t)$ será a derivada da Resposta ao Degrau

A resposta ao degrau $y_u(t)$ é a resposta homogênea $y_h(t)$ mais a resposta forçada $y_f(t)$

$$y_u(t) = y_h(t) + y_f(t)$$

1) Resposta Homogênea

$y'(t) + 6.25y(t) = 6.25\ (0)$

$y_h(t) = Ae^{-6.25t}$ Constante A será determinada no final com as condições iniciais.

Resposta Homogênea de 1ª Ordem

$y'(t) + \alpha\, y(t) = 0$

$y_h(t) = A\, e^{-\alpha t}$

Entrada Forçada	Resposta forçada
$f(t) = k_1$	$y_f(t) = B_1$
$f(t) = k_1 t + k_2$	$y_f(t) = B_1 t + B_2$
$f(t) = k_1 sen(w_1\, t)$	$y_f(t) = B_1 \cos(w_1\, t) + B_2 sen(w_1\, t)$
$f(t) = k_1 \cos(w_1\, t)$	$y_f(t) = B_1 \cos(w_1\, t) + B_2 sen(w_1\, t)$
$f(t) = k_1 e^{\beta t}$	$y_f(t) = B_1\, e^{\beta t}$
$f(t) = k_1 \cos(w_1\, t) + k_2 sen(w_1\, t)$	$y_f(t) = B_1 \cos(w_1\, t) + B_2 sen(w_1\, t)$

ct = 1/6.25=0.16

5*ct = 0.8s encerra o transitório

2) Resposta Forçada

$x(t) = 1\ para\ t \geq 0$

$y_f(t) = B$

Resposta Forçada tem a mesma natureza da Entrada Forçada

$y'(t) + 6.25y(t) = 6.25x(t)$

$(B\ )' + 6.25(B\ ) = 6.25\ (1)$

$0 + 6.25B = 6.25 \Rightarrow \quad B = 1$

$y_f(t) = 1$

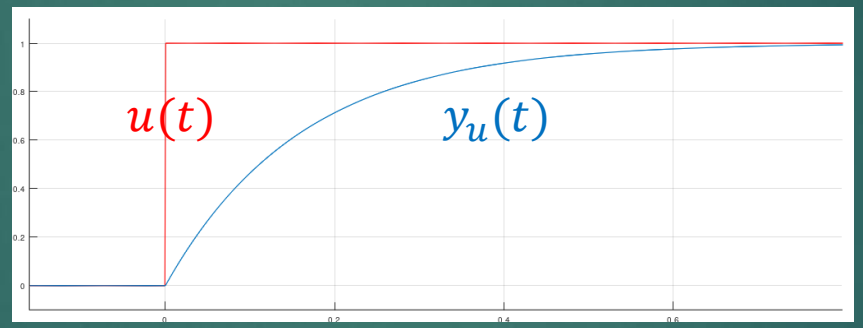
3) Resposta ao Degrau

$y_u(t) = y_h(t) + y_f(t) = Ae^{-6.25t} + 1$

Condições iniciais

$y_u(0) = 0 = Ae^{-6.25(0)} + 1 \Rightarrow A = -1$

Resposta ao degrau: $y_u(t) = (1 - e^{-6.25t})u(t)$



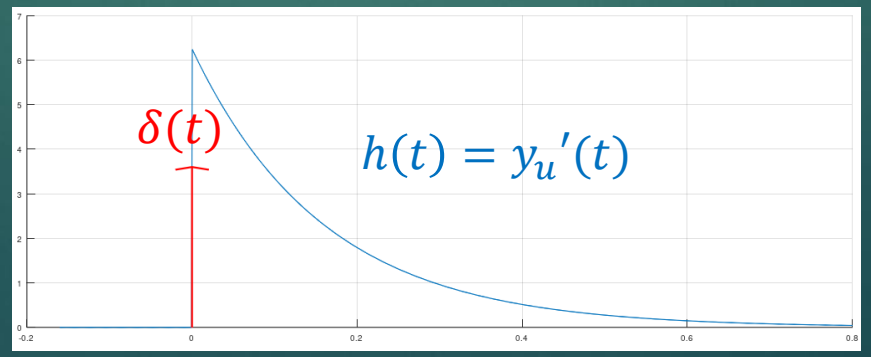

```
>> exemplo_A1()
```

```
>>    ct=1/6.25;
>>    t=linspace(-ct,5*ct,1000);
>>    yo = @(t) (1-exp(-6.25*t)).*f_u(t);
>>    hold on; grid on;
>>    plot(t,f_u(t),'r')
>>    plot(t,yo(t),'b')
>>    axis([-ct,5*ct])
```

4) Resposta ao Impulso

Resposta ao impulso é a derivada da resposta ao degrau

$h(t) = y_u{}'(t) = 6.25\, e^{-6.25t} u(t)$




```
>> exemplo_A2()
```

```
>>    ct=1/6.25;
>>    t=linspace(-ct,5*ct,3000);
>>    h = @(t)  6.25*exp(-6.25*t).*f_u(t);
>>    hold on; grid on;
>>    plot(t,h(t),"b")
>>    quiver(0,0,0,4, 'MaxHeadSize', 0.01, 'Color',
          'red', 'LineWidth', 1);
>>    axis([-ct,5*ct])
```

A resposta ao degrau $y_u(t)$ é a resposta homogênea $y_h(t)$ mais a resposta forçada $y_f(t)$

$$y_u(t) = y_h(t) + y_f(t)$$

Resposta Homogênea de 1ª Ordem

$$y'(t) + \alpha\, y(t) = 0$$

$$y_h(t) = A\, e^{-\alpha t}$$

Entrada Forçada	Resposta forçada
$f(t) = k_1$	$y_f(t) = B_1$
$f(t) = k_1 t + k_2$	$y_f(t) = B_1 t + B_2$
$f(t) = k_1 sen(w_1\, t)$	$y_f(t) = B_1 \cos(w_1\, t) + B_2 sen(w_1\, t)$
$f(t) = k_1 \cos(w_1\, t)$	$y_f(t) = B_1 \cos(w_1\, t) + B_2 sen(w_1\, t)$
$f(t) = k_1 e^{\beta t}$	$y_f(t) = B_1\, e^{\beta t}$
$f(t) = k_1 \cos(w_1\, t) + k_2 sen(w_1\, t)$	$y_f(t) = B_1 \cos(w_1\, t) + B_2 sen(w_1\, t)$

1) Resposta Homogênea

$$y'(t) + 6.25y(t) = 6.25\ (0)$$

$y_h(t) = Ae^{-6.25t}$ Constante A será determinada no final com as condições iniciais.

ct = 1/6.25=0.16

5*ct = 0.8s encerra o transitório

2) Resposta Forçada

$$x(t) = 10\cos(7t) \quad para\ t \geq 0$$

$$y_f(t) = B_1 \cos(7t) + B_2 \sin(7t)$$

Resposta Forçada tem a mesma natureza da Entrada Forçada

$$y'(t) + 6.25y(t) = 6.25x(t)$$

$$(B_1 \cos(7t) + B_2 \sin(7t)\ )' + 6.25(B_1 \cos(7t) + B_2 \sin(7t)\ ) = 6.25\ (10\cos(7t))$$

$$-7B_1 \sin(7t) + 7B_2 \cos(7t) + 6.25B_1 \cos(7t) + 6.25B_2 \sin(7t) = 62.5\cos(7t)$$

$$\begin{cases} 6.25B_1 + 7B_2 = 62.5 \\ -7B_1 + 6.25B_2 = 0 \end{cases}$$

$$y_f(t) = 4.44\cos(7t) + 4.97\sin(7t)$$

3) Resposta ao Degrau

$$y_u(t) = y_h(t) + y_f(t) = Ae^{-6.25t} + 4.44\cos(7t) + 4.97\sin(7t)$$

Condições iniciais

$$y_u(0) = 5 = A + 4.44 + (0)$$

$$\Rightarrow A = 0.56$$

Resposta ao degrau: $y_u(t) = (0.56e^{-6.25t} + 4.44\cos(7t) + 4.97\sin(7t)\ )u(t)$

```
>> exemplo_A1()
```

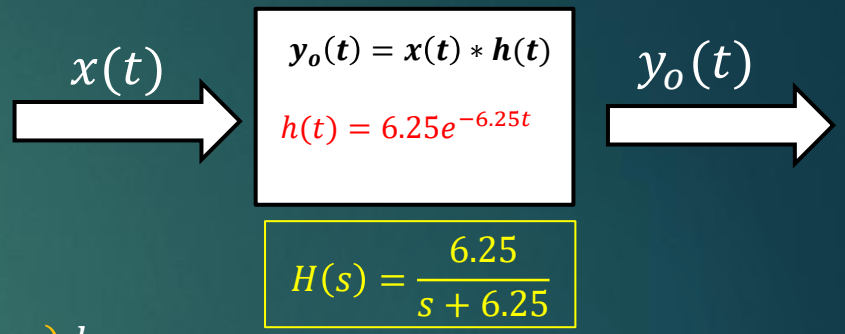
```
>>    ct=1/6.25;
>>    t=linspace(-ct,5*ct,1000);
>>    yo = @(t) (1-exp(-6.25*t)).*f_u(t);
>>    hold on; grid on;
>>    plot(t,f_u(t),'r')
>>    plot(t,yo(t),'b')
>>    axis([-ct,5*ct])
```

```
>> exemplo_A2()
```

```
>>    ct=1/6.25;
>>    t=linspace(-ct,5*ct,3000);
>>    h = @(t)  6.25*exp(-6.25*t).*f_u(t);
>>    hold on; grid on;
>>    plot(t,h(t),"b")
>>    quiver(0,0,0,4, 'MaxHeadSize', 0.01, 'Color',
          'red', 'LineWidth', 1);
>>    axis([-ct,5*ct])
```

Calcular por Convolução a Resposta ao Estado Nulo $y_o(t)$ deste sistema para a entrada $x(t) = 10.5\, e^{-t}u(t)$

Como já conhecemos a resposta ao impulso $h(t) = 6.25\, e^{-6.25t}\mathrm{u(t)}$, a resposta do sistema ao sinal $10.5e^{-t}$ será simplesmente a convolução $x(\mathrm{t})$ com $h(t)$



$$H(s) = \frac{6.25}{s + 6.25}$$

$$y_o(\mathrm{t}) = \mathrm{x(t)} * \mathrm{h(t)}$$

$$y_o(\mathrm{t}) = [10.5e^{-t}u(t)] * [6.25\, e^{-6.25t}\mathrm{u(t)}]$$

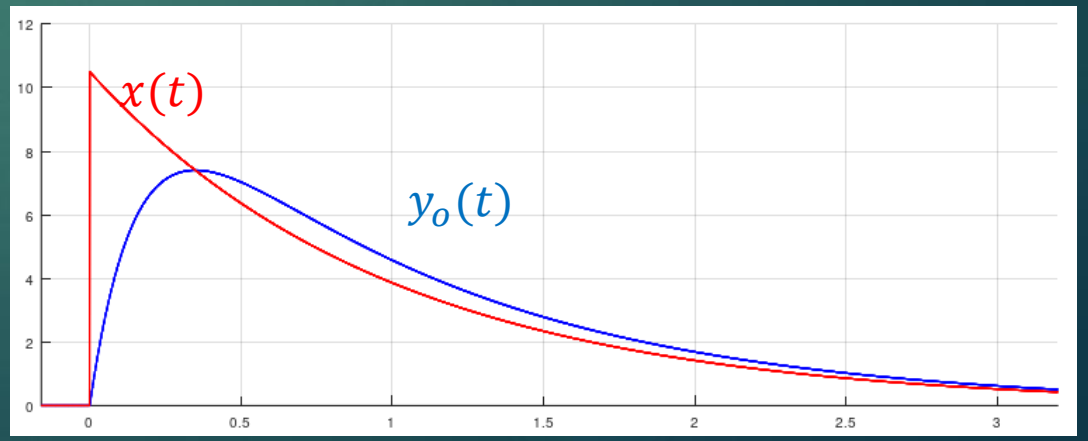
$$y_0(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{\infty} 10.5e^{-(\tau)}u(\tau)6.25e^{-6.25(t-\tau)}u(t-\tau)d\tau$$

$$y_o(t) = 65.625\int_0^t e^{-(\tau)}e^{-6.25(t-\tau)}d\tau = 65.625\int_0^t e^{-6.25t+5.25\tau}d\tau = 12.5\, e^{-6.25t+5.25\tau}\Big|_0^t$$

$$y_o(t) = 12.5e^{-t} - 12.5\, e^{-6.25t}$$

```
>>      ct=1/6.25;
>>      t=linspace(-ct,20*ct,3000);
>>      xi = @(t)  10.5*exp(-t).*f_u(t);
>>      yo = @(t) (12.5*exp(-t)-12.5*exp(-6.25*t)).*f_u(t);
>>      hold on; grid on;
>>      plot(t,yo(t),"b");
>>      plot(t,xi(t),"r");
>>      axis([-ct,20*ct])
```

```
>> exemplo_A3o()
```



Calcular por Convolução a Resposta ao Estado Nulo $y_o(t)$ deste sistema para a entrada $x(t) = 10\cos(7t)u(t)$

Como já conhecemos a resposta ao impulso $h(t) = 6.25\, e^{-6.25t}\mathrm{u(t)}$, a resposta do sistema ao sinal $x(\mathrm{t}) = 10\cos(7t)u(t)$ será simplesmente a convolução $x(\mathrm{t})$ com $h(t)$

$$x(t) \longrightarrow \boxed{\begin{array}{l} \boldsymbol{y_o(t) = x(t) * h(t)} \\ h(t) = 6.25e^{-6.25t} \end{array}} \longrightarrow y_o(t)$$

$$\boxed{H(s) = \frac{6.25}{s + 6.25}}$$

$$y_o(\mathrm{t}) = \mathrm{x(t)} * \mathrm{h(t)}$$

$$y_o(\mathrm{t}) = [10\cos(7t)u(t)] * [6.25\, e^{-6.25t}\mathrm{u(t)}]$$

$$y_0(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{\infty} 10cos(7\tau)u(\tau)6.25e^{-6.25(t-\tau)}u(t-\tau)d\tau$$

$$y_o(t) = 62.5\int_0^t cos(7\tau)e^{-6.25(t-\tau)}d\tau = \left. -4.44\, e^{-6.25(t-\tau)}\cos(7\tau) + 4.97\, e^{-6.25(t-\tau)}\sin(7\tau)\right|_0^t$$

$$y_o(t) = 4.44\, e^{-6.25(t-t)}\cos(7t) + 4.97\, e^{-6.25(t-t)}\sin(7\mathrm{t}) - \left(4.44\, e^{6.25(t-0)}\cos(7(0)) + 4.97\, e^{-6.25(t-0)}\sin(7(0))\right)$$

$$y_o(t) = 4.44\cos(7t) + 4.97\sin(7\mathrm{t}) - 4.44\, e^{-6.25t}$$

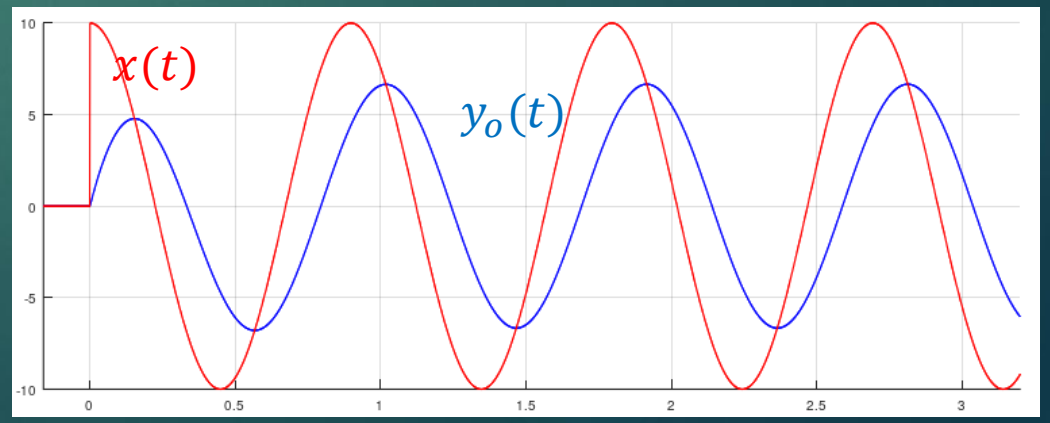
$$y_o(t) = -4.44\, e^{-6.25t} + \sqrt{4.44^2 + 4.97^2}\cos\left(7t - \mathrm{atan}\left(\frac{4.97}{4.44}\right)\right)$$

$$y_o(\mathrm{t}) = [-4.44\mathrm{e}^{-6.25\mathrm{t}} + 6.66\cos(7\mathrm{t} - 0.842)]\mathrm{u(t)}$$

$$\boxed{\begin{array}{l} A\cos(7t) + B\sin(7\mathrm{t}) = \sqrt{A^2 + B^2}\sin(7t + \mathrm{atan}(A,B)) \\ A\cos(7t) + B\sin(7\mathrm{t}) = \sqrt{A^2 + B^2}\cos(7t - \mathrm{atan}(B,A)) \end{array}}$$

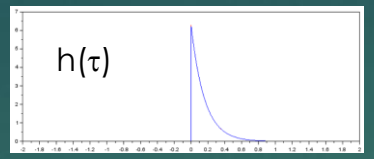
```
>> exemplo_A4o()
```

```
>>    ct=1/6.25;
>>    t=linspace(-ct,20*ct,3000);
>>    xi = @(t) 10*cos(7*t).*f_u(t);
>>    yo = @(t) (-4.44*exp(-6.25*t)+6.66*cos(7*t-0.842)).*f_u(t);
>>     hold on; grid on;
>>     plot(t,yo(t),"b");
>>     plot(t,xi(t),"r");
>>     axis([-ct,20*ct]);
```

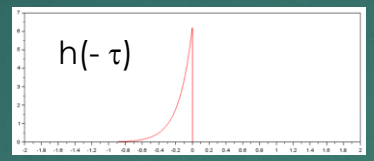


Interpretação Geométrica da Convolução

1. Inverter h(τ) → h(-τ)
2. Deslocar h(-τ) → h(t-τ)
3. Multiplicar x(τ) por h(t-τ) → x(τ) h(t-τ)
4. "Somar" – integrar para todo τ



$$x(t) = 10\cos(7t)u(t)$$
$$h(t) = 6.25\, e^{-6.25t}\mathrm{u(t)}$$
$$y_o(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$



$-\infty \le t < 0$

$y_o(t) = 0$ *(não há sobreposição)*

$t \ge 0$

$$y_o(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{\infty} 10cos(7\tau)u(\tau)6.25e^{-6.25(t-\tau)}u(t-\tau)d\tau$$

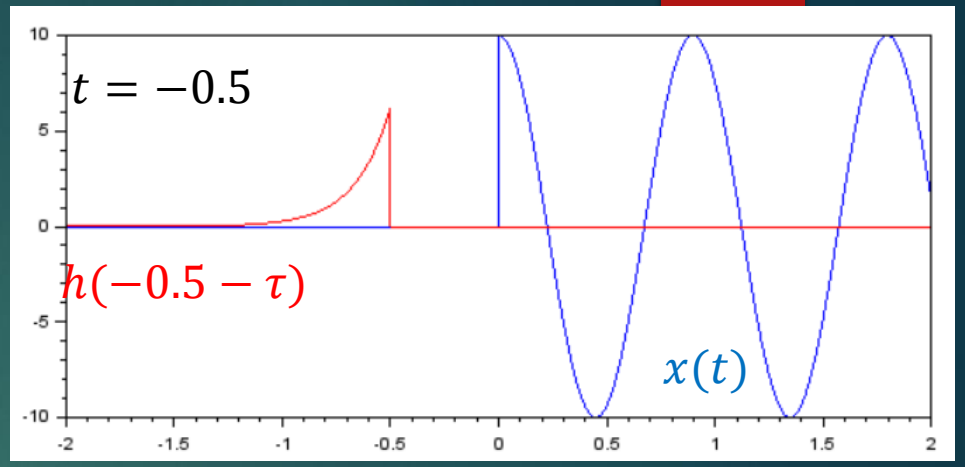
$$y_o(t) = 62.5\int_0^t cos(7\tau)e^{-6.25(t-\tau)}d\tau$$

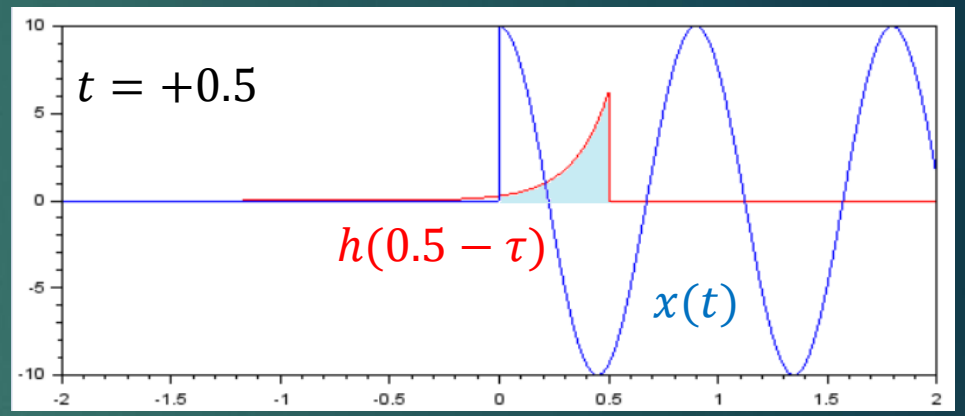
(sobreposição entre $0 \le \tau \le t$*)*

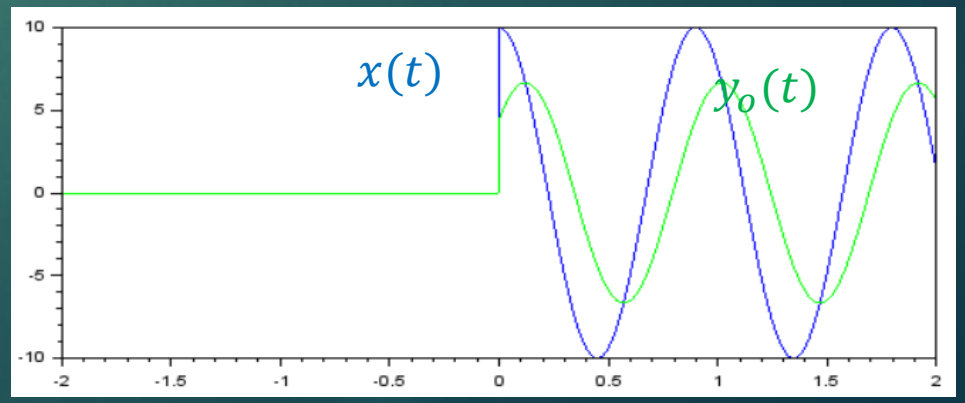
$$y_o(t) = -4.44\, e^{-6.25(t-\tau)}\cos(7\tau) + 4.97\, e^{-6.25(t-\tau)}\sin(7\tau)\Big|_0^t$$

$$y_o(t) = 4.44\ \cos(7t) + 4.97\sin(7t) - 4.44\, e^{-6.25t}$$

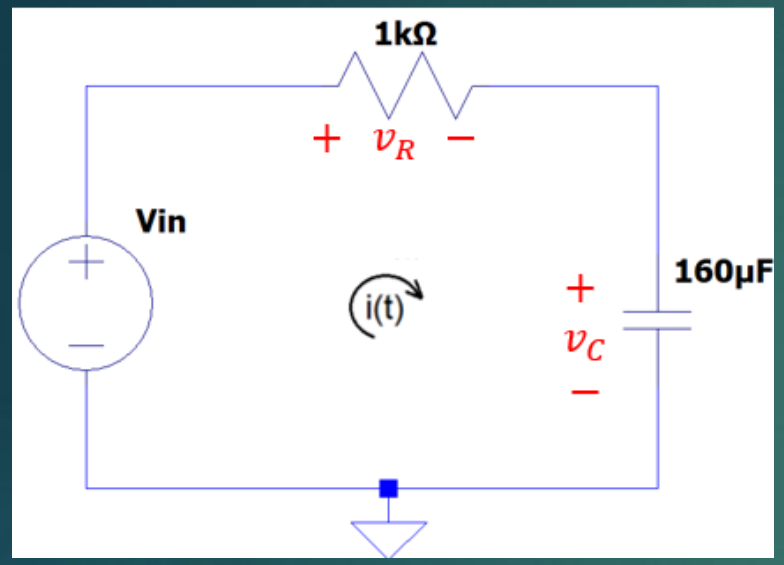
$$y_o(\mathrm{t}) = [6.66\cos(7\mathrm{t} - 0.842) - 4.44\mathrm{e}^{-6.25\mathrm{t}}]\mathrm{u(t)}$$

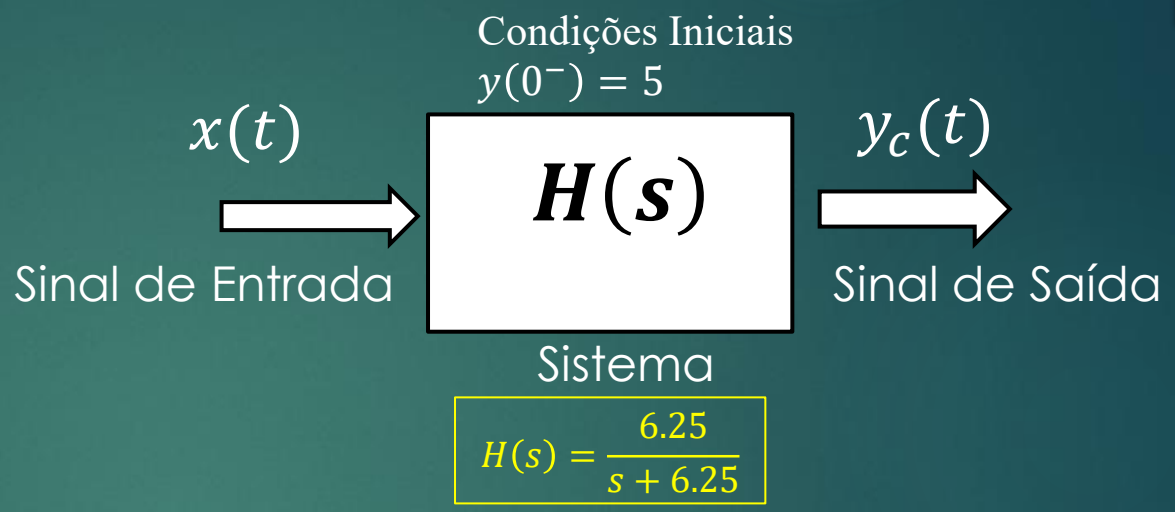






Calcular a Resposta Completa $y_c(t)$ deste sistema para entrada $x(t) = 10\cos(7t)u(t)$, com **condições iniciais** $y_c(0^-) = 5V$, **mas agora com condições iniciais**





- Calcular $y(t)$ *para* $x(t)$
- $x(t) = 10\cos(7t)u(t)$
- Condições Iniciais
- $y(0^-) = 5$

Equação diferencial:

- $y'(t) + 6.25y(t) = 6.25x(t)$

Com condições iniciais, precisamos somente somar a **Resposta à Entrada Nula** $y_n(t)$ à **Resposta ao Estado Nulo** $y_0(t)$ que já foi calculada

$y_c(t) = y_0(t) + y_n(t)$

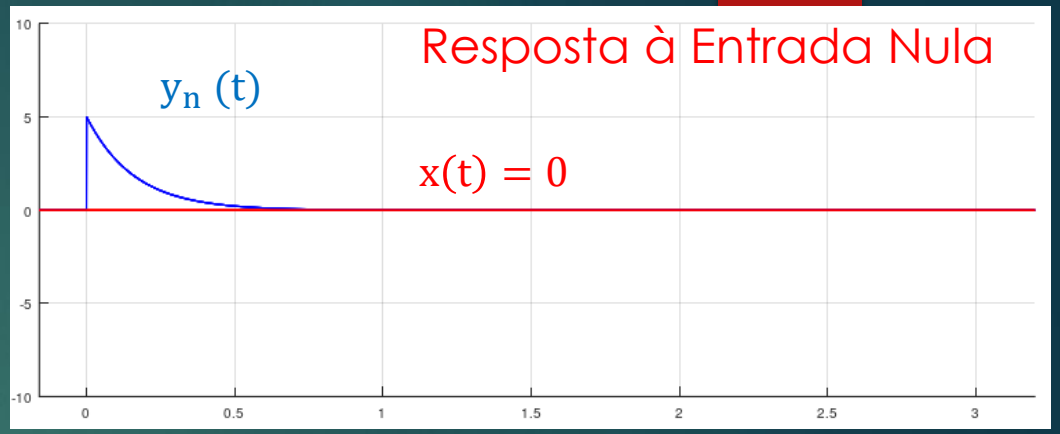
$y_n(t)$ é semelhante á reposta homogênea, incluindo as condições iniciais

4) Resposta à Entrada Nula $y_n(t)$ é semelhante á reposta homogênea, só que com condições iniciais

$y_n(t) = Ae^{-6.25t}$
Condição inicial $y_n(0) = 5$
$y_n(0) = 5 = Ae^{-0}$
$A = 5$
$y_n(\mathrm{t}) = [5\ \mathrm{e}^{-6.25\mathrm{t}}]\mathrm{u}(\mathrm{t})$

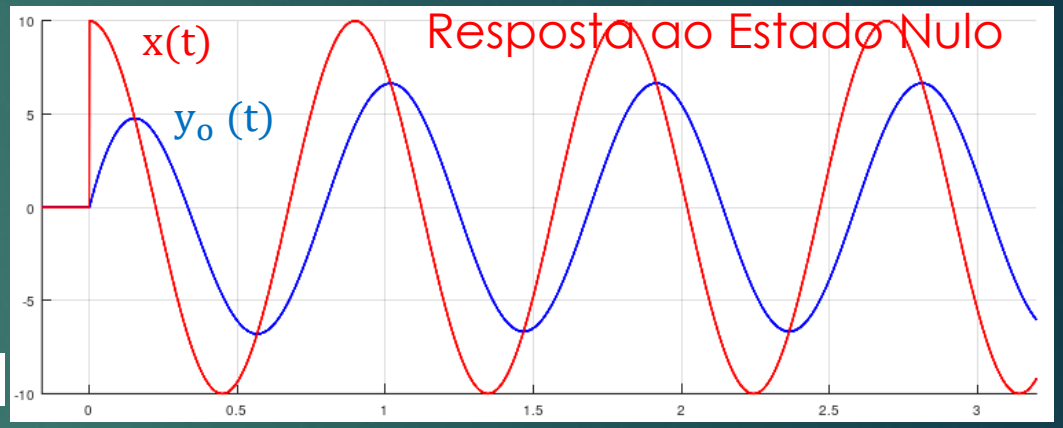
Resposta Homogênea de 1ª Ordem
$y'(t) + \alpha\, y(t) = 0$
$y_h(t) = A\, e^{-\alpha t}$

```
--> exemplo_A4n()
```



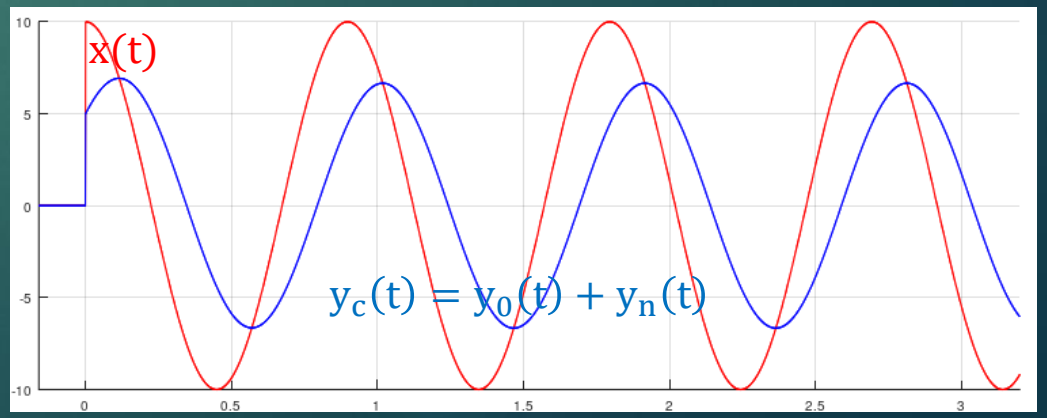
5) A Resposta Completa $y_c(t)$ será a soma da Resposta ao Estado Nulo $y_0(t)$ (já calculada) com a Resposta à Entrada Nula $y_n(t)$

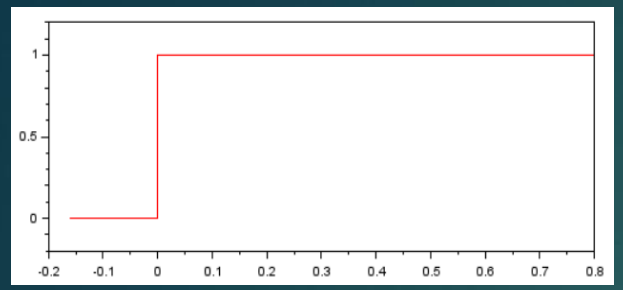
```
--> exemplo_A4o()
```



$y_c(t) = y_{o(t)} + y_n(t)$
$y_c(t) = \left(-4.44\mathrm{e}^{-6.25\mathrm{t}} + 6.66\cos(7\mathrm{t} - 0.842)\right) + (5\ \mathrm{e}^{-6.25\mathrm{t}})$
$y_c(t) = 0.56\ \mathrm{e}^{-6.25\mathrm{t}} + 6.66\cos(7\mathrm{t} - 0.842)$

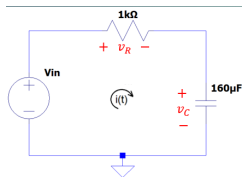
```
--> exemplo_A4c()
```





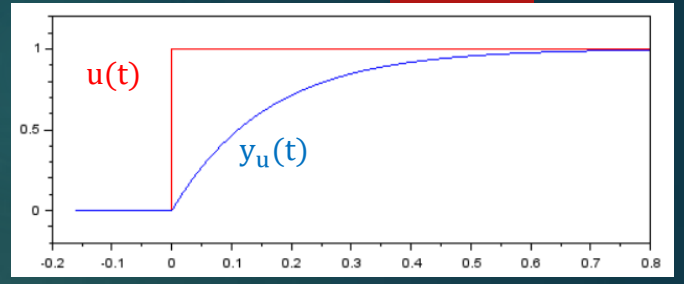
$x(t) = u(t)$

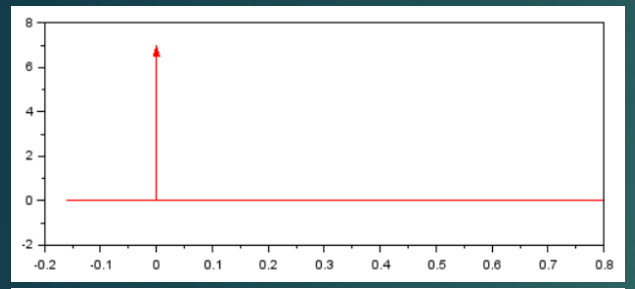
$$y'(t) + 6.25y(t) = 6.25x(t)$$



$y(0^-) = 0$

$y_u(t) = 1 - e^{-6.25t}$





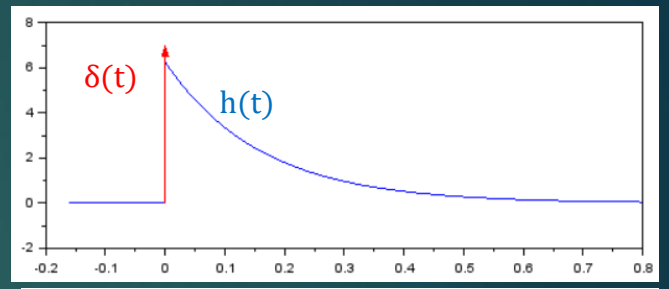
$x(t) = \delta(t)$

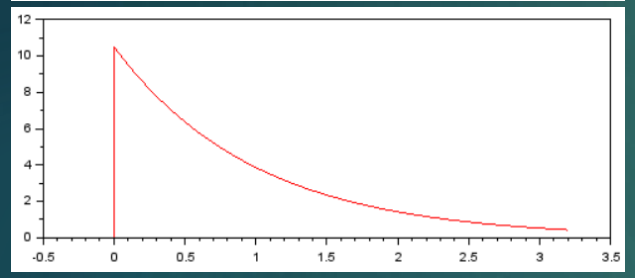
$$y'(t) + 6.25y(t) = 6.25x(t)$$

$h(t) = y_u{}'(t)$

$y(0^-) = 0$

$h(t) = 6.25\, e^{-6.25t}$



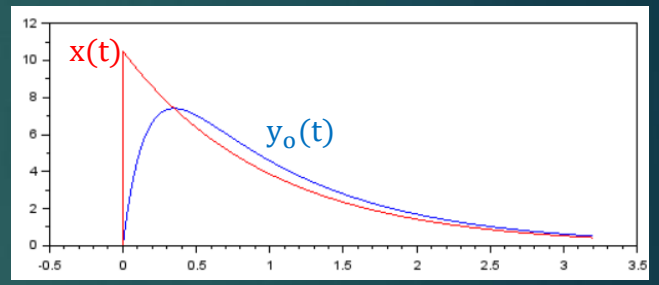


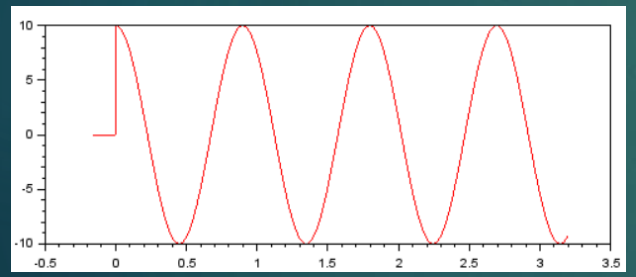
$x(t) = 10.5\, e^{-t}$

$h(t) = 6.25\, e^{-6.25t}$

$y_o(t) = x(t) * h(t)$

$y_o(t)$
$= -12.5\, e^{-6.25t} + 12.5e^{-t}$



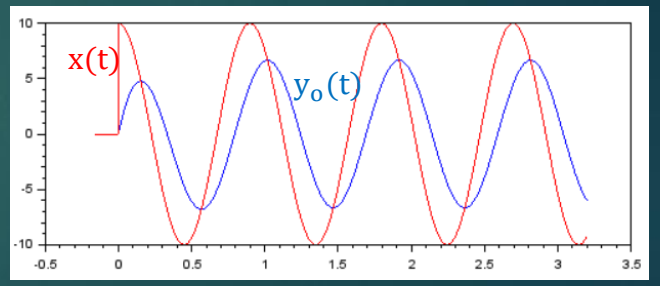


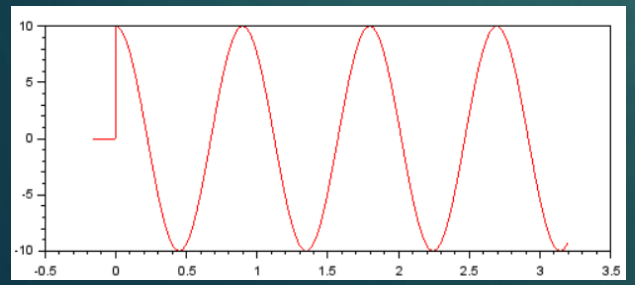
$x(t) = 10\cos(7t)$

$h(t) = 6.25\, e^{-6.25t}$

$y_o(t) = x(t) * h(t)$

$y_o(t) = -\,4.44e^{-6.25t}$
$+6.66\cos(7t - 0.842)$





$x(t) = 10\cos(7t)$

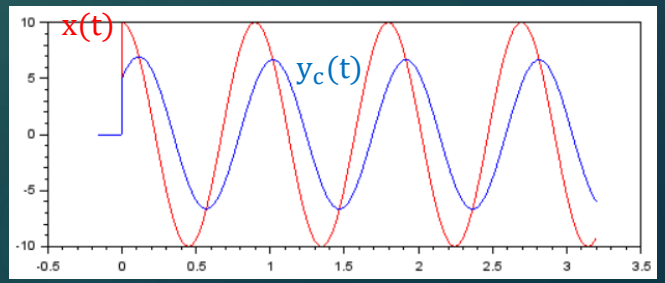
$h(t) = 6.25\, e^{-6.25t}$

$y_o(t) = x(t) * h(t)$

$y_c(t) = y_{o(t)} + y_n(t)$

$y(0^-) = 5$

$y_c(t) = 0.56\, e^{-6.25t}$
$+6.66\cos(7t - 0.842)$

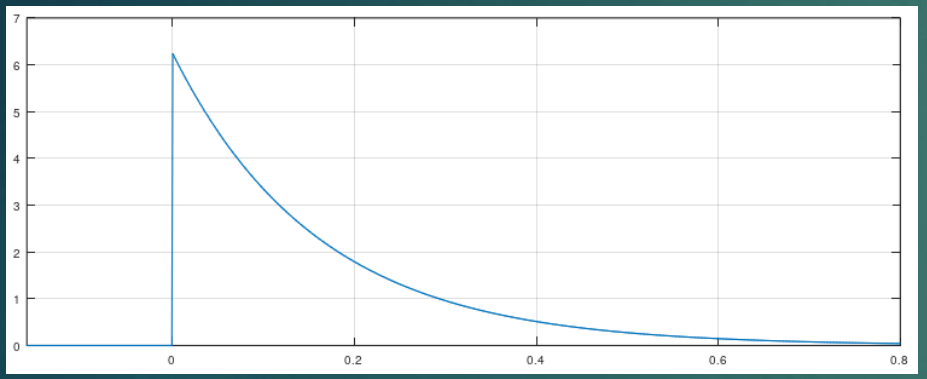


Aproximação de Equações Diferenciais por Equações de Diferenças

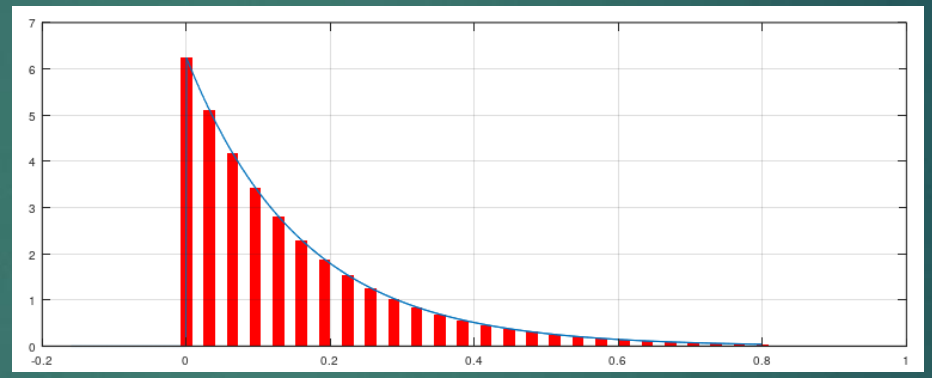
Seja sinal contínuo $y(t)$ no intervalo [a,b], da qual obtemos N amostras $y(n)$, com índices de n=[0:N-1], espaçadas de T segundos. As amostras $y(n)$ aproximam o sinal analógico $y(t)$ no intervalo.

$$y(t) \cong y(nT)$$

```
>> h = @(t) (6.25*exp(-6.25*t)).*f_u(t);
>> ct=1/6.25;
>> ini=-ct;
>> fim=5*ct;
>> t=linspace(ini,fim,1001);
>> plot(t,h(t))
```




```
>> T=ct/5;
>> ts=[ini:T:fim];
>> hold on; grid on;
>> bar(ts,h(ts),0.5,'r','edgecolor','none')
>> plot(t,h(t))
```




```
>> n=floor(ts/T);
>> [n' ts' h(ts)']
ans =

          -5        -0.16            0
          -4       -0.128            0
          -3       -0.096            0
          -2       -0.064            0
          -1       -0.032            0
           0            0         6.25
           1        0.032       5.1171
           2        0.064       4.1895
           3        0.096       3.4301
           4        0.128       2.8083
           5         0.16       2.2992
           5        0.192       1.8825
           7        0.224       1.5412
           8        0.256       1.2619
           9        0.288       1.0331
           9         0.32      0.84585
          11        0.352      0.69252
          12        0.384      0.56699
          13        0.416      0.46421
          13        0.448      0.38006
          15         0.48      0.31117
          16        0.512      0.25476
          16        0.544      0.20858
          18        0.576      0.17077
          19        0.608      0.13982
          20         0.64      0.11447
          21        0.672     0.093722
          22        0.704     0.076733
          23        0.736     0.062824
          24        0.768     0.051436
          24          0.8     0.042112
```

Podemos também aproximar as derivadas de $y(t)$ a partir das amostra $y[n]$:

$$y(t) \cong y[n]$$

$$y'(t) \cong \frac{y[n] - y[n-1]}{T}$$

$$y''(t) \cong \frac{y[n] - 2y[n-1] + y[n-2]}{T^2}$$

Equação de Recorrência (Diferenças) a partir da Equação Diferencial

Transformar um sistema analógico em digital

$$y'(t) + 6.25y(t) = 6.25x(t)$$

$$y(t) \cong y[n]$$
$$y'(t) \cong \frac{y[n] - y[n-1]}{T}$$
$$x(t) \cong x[n]$$

Diferença Finitas

$$\frac{y[n] - y[n-1]}{T} + 6.25y[n] = 6.25\, x[n]$$

$$y[n] - y[n-1] + 6.25\, T\, y[n] = 6.25\, T\, x[n]$$

$$(1 + 6.25T)y[n] - y[n-1] = 6.25\, T\, x[n]$$

$$y[n] = \frac{1}{(1 + 6.25\, T)} y[n-1] + \frac{6.25T}{(1 + 6.25\, T)} x[n]$$

$$T = \frac{c_t}{5} = 0.032$$

$$y[n] = \frac{1}{(1 + 6.25(0.032))} y(n-1) + \frac{6.25(0.032)}{(1 + 6.250(0.032))} x[n]$$

$$y[n] = 5/6\, y[n-1] + 1/6\, x[n]$$

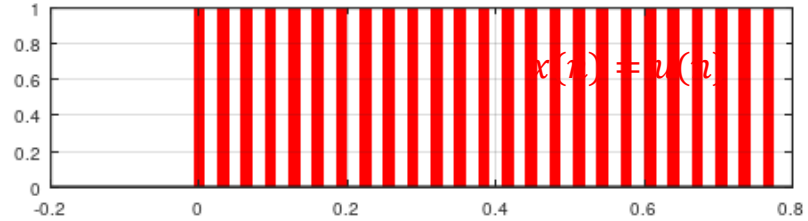
A Escolha de T (período de Amostragem) depende do conhecimento do teorema da amostragem (Nyquist) e dos polos e zeros do Sistema.

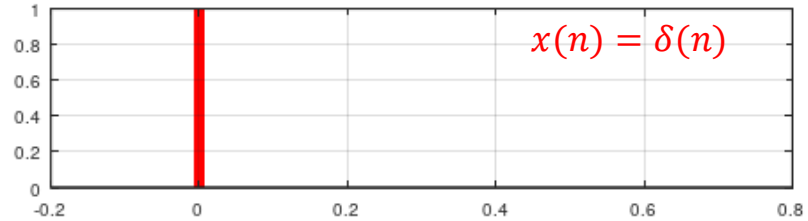
Como simplificação, vamos obter arbitrariamente 5 amostras em uma constante de tempo ct da equação diferencial.

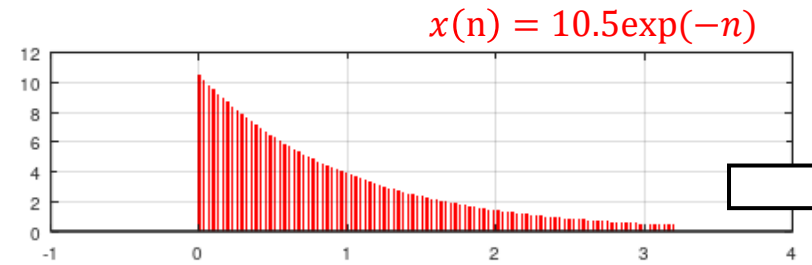
$$\mathrm{h(t)} = 6.25\, e^{-6.25t}$$

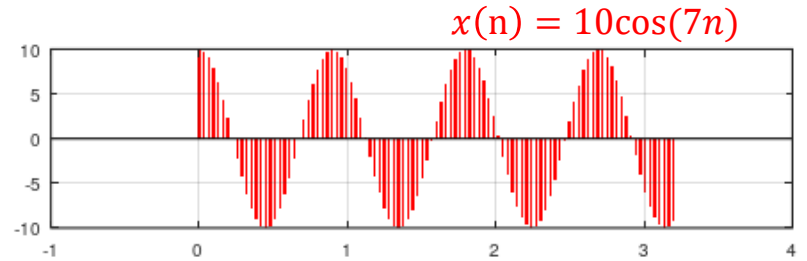
$$c_t = \frac{1}{6.25} = 0.16s \qquad T = \frac{ct}{5} = 0.032s$$

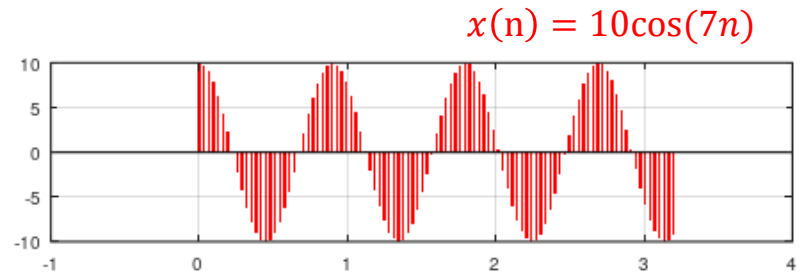
```
function y_out=Sistema_Eletrico_1aOrdem(x_in,y_ini)
    xi=[0 x_in];
    yi=[y_ini];
    for (i=2:length(xi))
        yi(i)=(1/6)*xi(i)+(5/6)*yi(i-1);
    end
    y_out=yi(2:length(xi));
endfunction
```











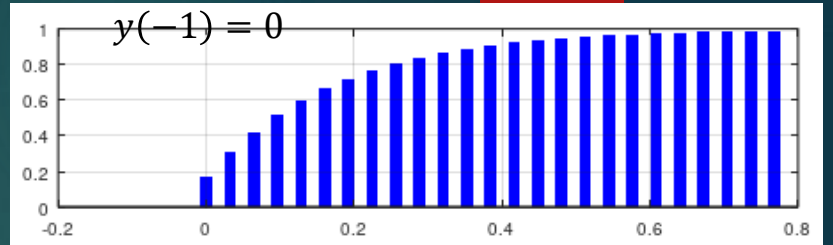
$$T = 0.032$$

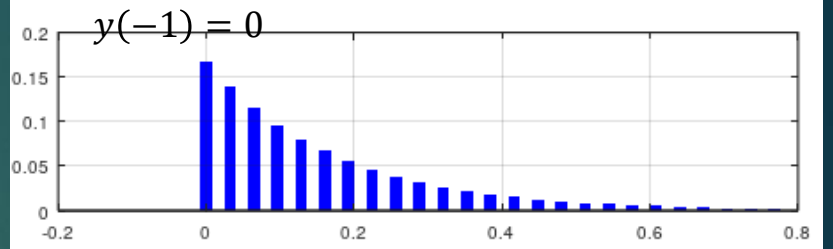
```
--> exemplo_A5(0.032)
```

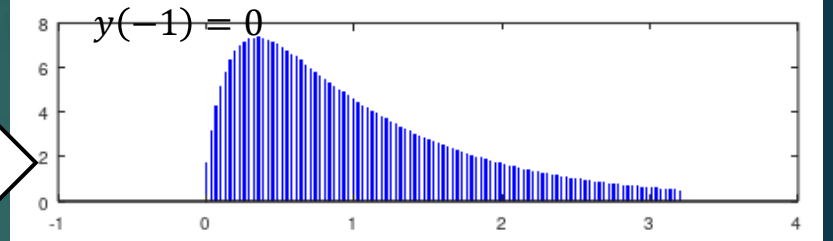
$$y[n] = \frac{1}{(1 + 6.25(0.032))} y[n-1] + \frac{6.25(0.032)}{(1 + 6.250(0.032))} x[n]$$

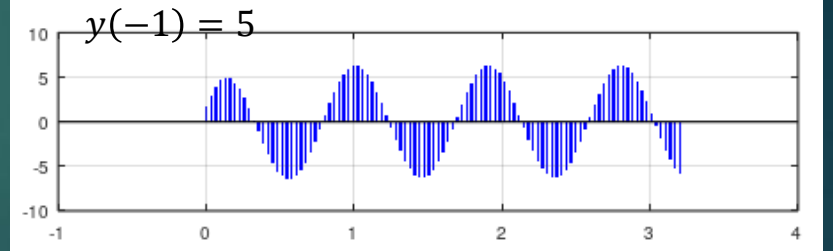
$$y[n] = 5/6\, y[n-1] + 1/6\, x[n]$$

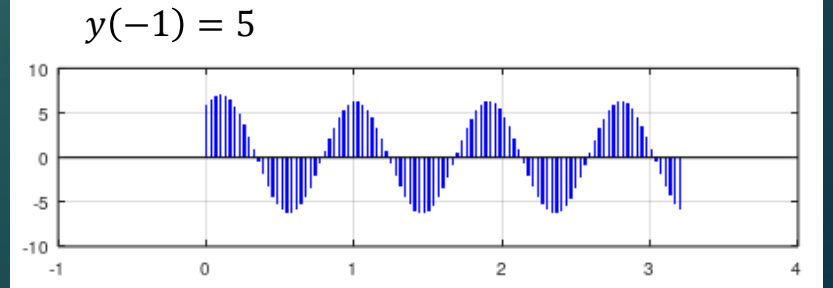
```
function y_out=Sistema_Eletrico_1aOrdem(x_in,y_ini)
    xi=[0 x_in];
    yi=[y_ini];
    for (i=2:length(xi))
        yi(i)=(1/6)*xi(i)+(5/6)*yi(i-1);
    end
    y_out=yi(2:length(xi));
endfunction
```

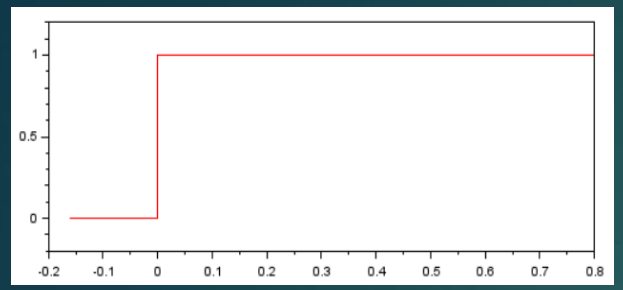






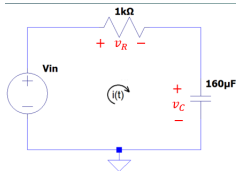






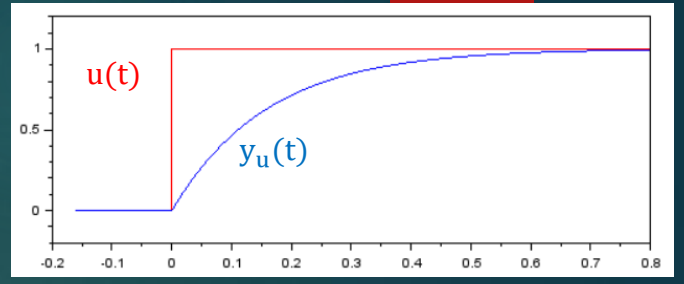
$x(t) = u(t)$

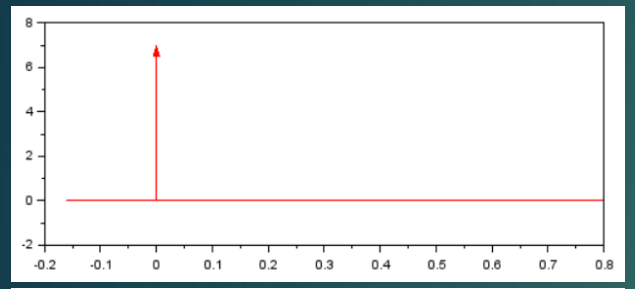
$y'(t) + 6.25y(t) = 6.25x(t)$



$y(0^-) = 0$

$y_u(t) = 1 - e^{-6.25t}$





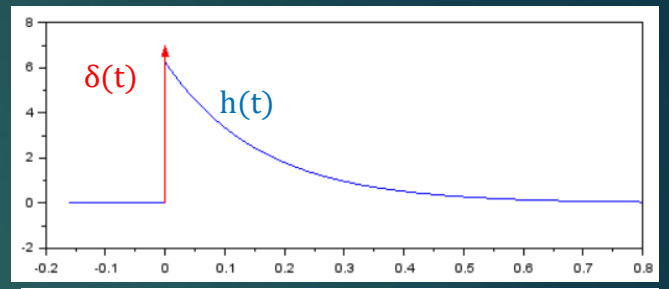
$x(t) = \delta(t)$

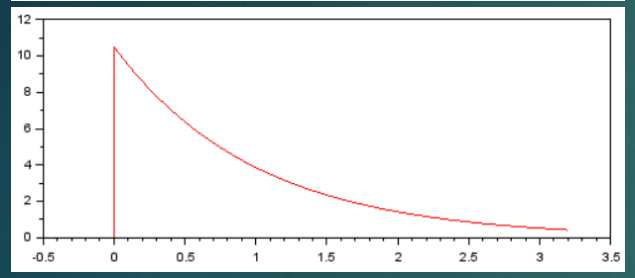
$y'(t) + 6.25y(t) = 6.25x(t)$

$h(t) = y_u{}'(t)$

$y(0^-) = 0$

$h(t) = 6.25\, e^{-6.25t}$



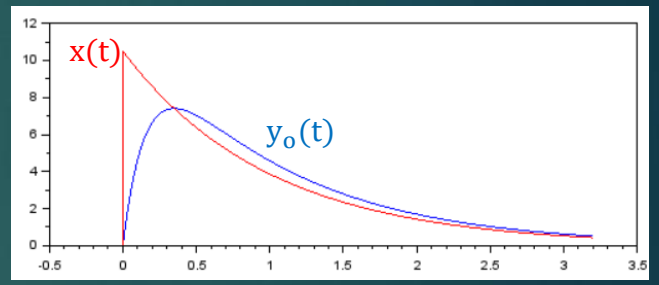


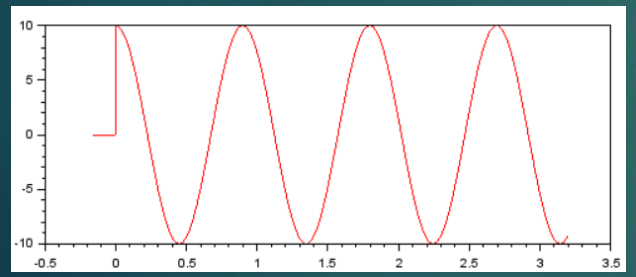
$x(t) = 10.5\, e^{-t}$

$h(t) = 6.25\, e^{-6.25t}$

$y_o(t) = x(t) * h(t)$

$y_o(t)$
$= -12.5\, e^{-6.25t} + 12.5e^{-t}$



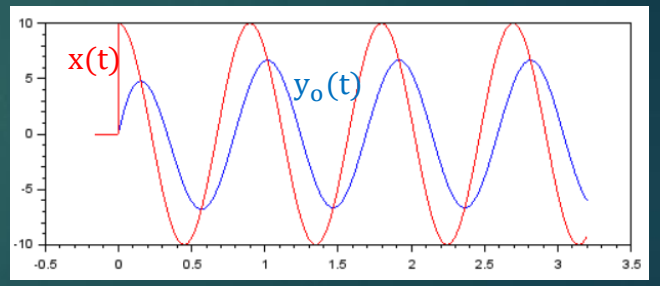


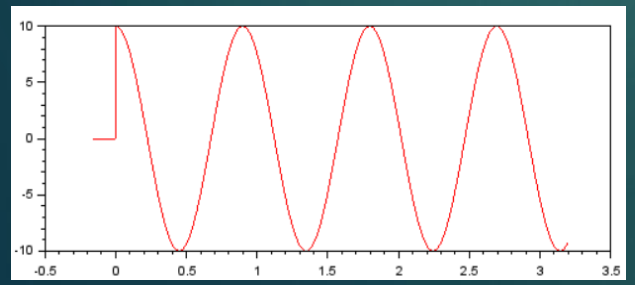
$x(t) = 10\cos(7t)$

$h(t) = 6.25\, e^{-6.25t}$

$y_o(t) = x(t) * h(t)$

$y_o(t) = -\,4.44e^{-6.25t}$
$+6.66\cos(7t - 0.842)$





$x(t) = 10\cos(7t)$

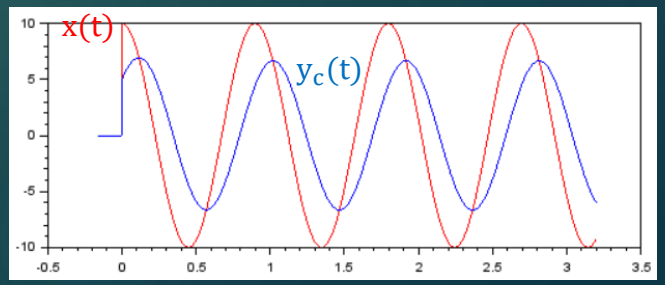
$h(t) = 6.25\, e^{-6.25t}$

$y_o(t) = x(t) * h(t)$

$y_c(t) = y_{o(t)} + y_n(t)$

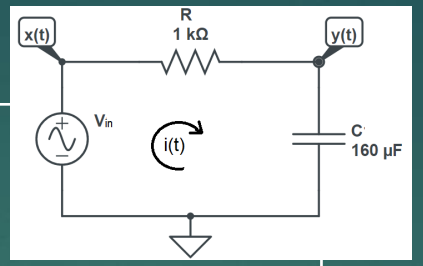
$y(0^-) = 5$

$y_c(t) = 0.56\, e^{-6.25t}$
$+6.66\cos(7t - 0.842)$



a) Analisamos a equação diferencial e encontramos a sua resposta ao Degrau $y_u(t)$ de depois a Resposta ao Impulso $h(t)$

$$\boxed{y'(t) + \mathbf{6.25}y(t) = 6.25x(t)}$$



1) $y_o(t) = y_h(t) + y_f(t)$

1.1) $y_h(t)$ de 1ª ordem tem a forma $A_1 e^{-6.25t}$

A_1 é avaliada em (1.3)

1.2) $y_f(t) = B$

Encontrar B´s substituindo $y_f(t)$ na eq. diferencial

1.3) $y_u(t) = A_1 e^{-6.25t} + B = 1 - e^{-6.25t}$

Encontrar A_1 substituindo as condições iniciais nulas

1.4) $h(t) = y'_u(t) = 6.25\, e^{-6.25t}$

Resposta Homogênea de 1ª Ordem

$y'(t) + \alpha\, y(t) = 0$

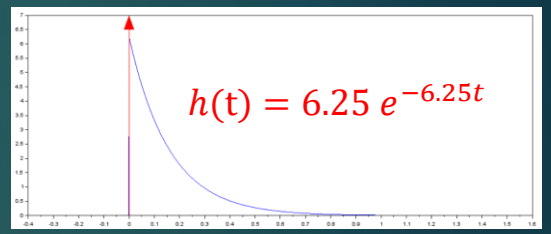
$y_h(t) = A\, e^{-\alpha t}$

Resposta Forçada

$x(t) = 1 \qquad y_f(t) = B$

b) Encontramos a resposta ao estado nulo $y_o(t)$ a uma entrada genérica $x(t)$ por convolução

$x(t)$ → $\boxed{H(s) = \dfrac{6.25}{s + 6.25}}$ → $y_o(t) = x(t) * h(t) = x(t) * (6.25\, e^{-6.25t})$



d) Se houver condições iniciais, somamos a resposta à Entrada Nula $y_n(t)$ à resposta ao Estado Nulo $y_o(t)$

$y_n(t)$ tem forma semelhante à resposta homogênea

$y_n(t) = A_2\, e^{-6.25t}$ Encontrar A_2 substituindo as condições iniciais

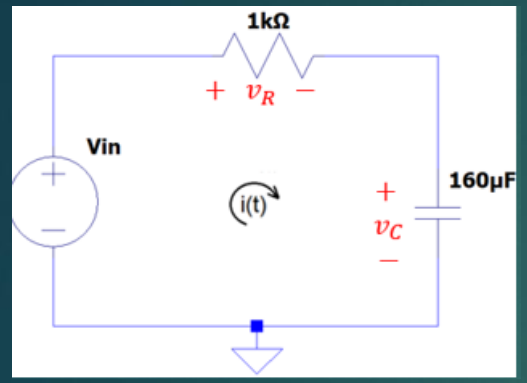
$y_c(t) = y_0(t) + y_n(t)$ A Resposta completa

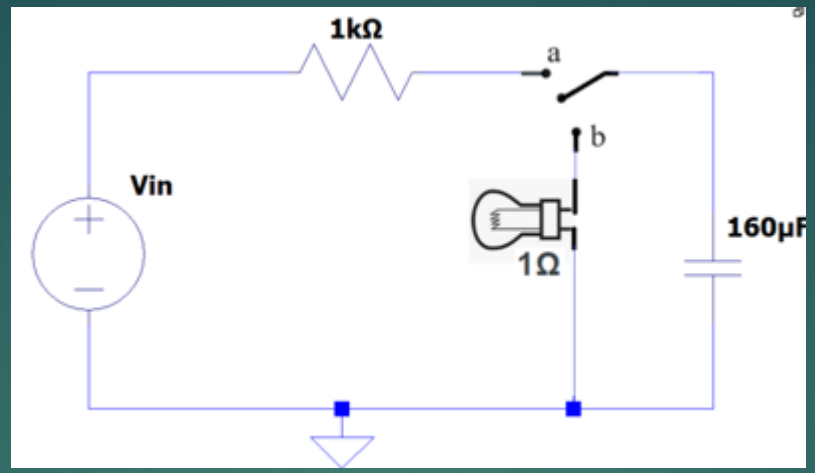
e) Encontramos a saída completa discreta $y(n)$ transformado a equação diferencial em uma equação de diferenças (algoritmo). Para isso tivemos que escolher um período de amostragem T, para transforma a função contínua em discreta.

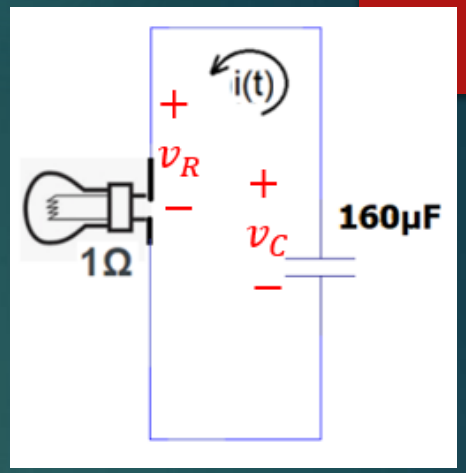
$$\boxed{y'(t) + \mathbf{6.25}y(t) = 6.25x(t)} \rightarrow \boxed{y[n] = 5/6\, y[n-1] + 1/6\, x[n]}$$

$$\boxed{H(s) = \frac{6.25}{s + 6.25}} \qquad \boxed{H(z) = \frac{(1/6)}{1 - \frac{5}{6}z^{-1}}}$$

Voltando ao Flash da máquina fotográfica







$$v_c'(t) + \frac{1}{\mathrm{RC}} v_c(t) = \frac{1}{RC} v_{in}(t)$$

$RC = 0.16\ s = 1/6.25;\ v_{in} = 300\ V;$

$y'(t) + \mathbf{6.25} y(t) = 6.25 * 300\ \ ; v_C(0^-) = 0$

$$v_c'(t) + \frac{1}{\mathrm{RC}} v_c(t) = 0$$

$RC = 0.00016\ s = 1/6250\ ;$

$y'(t) + \mathbf{6250} y(t) = 0;$

$y_n(t) = Ae^{-6250t}$

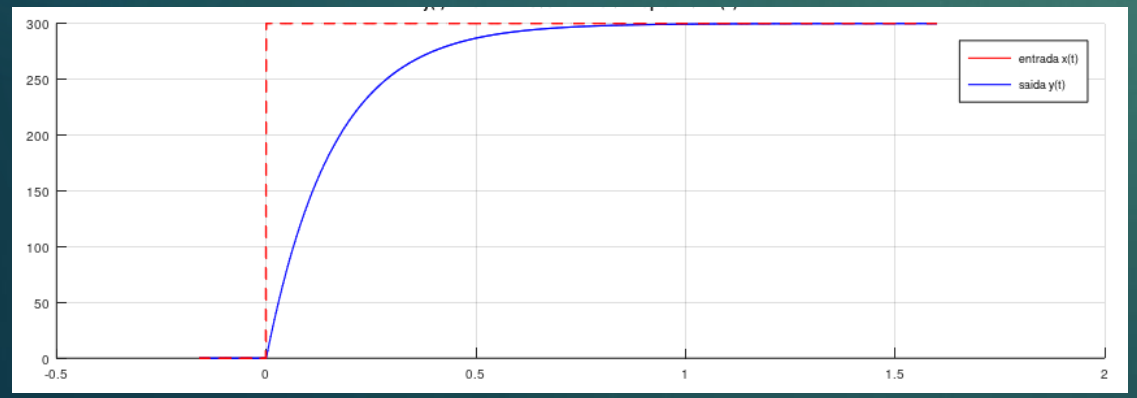
Condição inicial

$y_n(0^-) = 300 = Ae^{-0}$

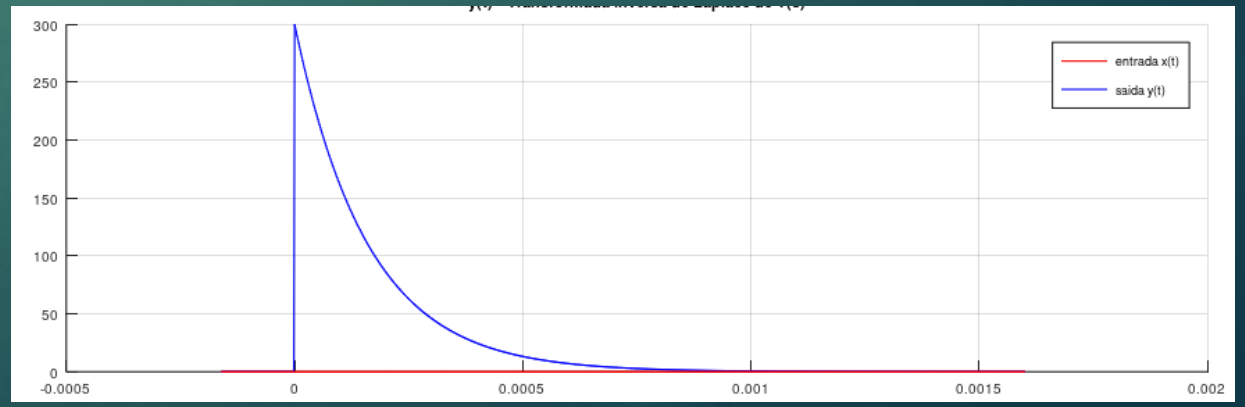
$A = 300$

$y_n(\mathrm{t}) = [300\ \mathrm{e}^{-625\mathrm{t}}]\mathrm{u}(\mathrm{t})$

$$y(t) = 300(1 - e^{-6.25t})$$



$$y = 300e^{-6250t}$$ (Somente Entrada Nula)



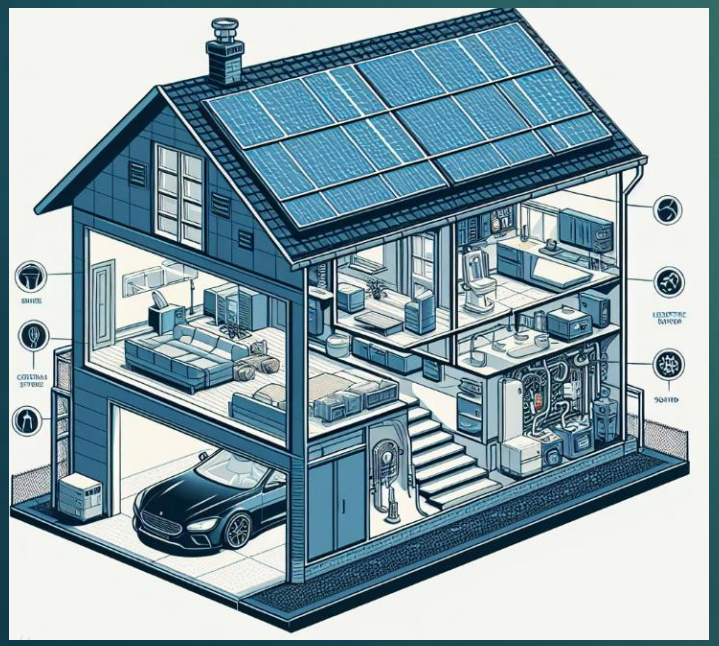
B) Resposta de Sistemas Eletrônicos RLC de 2ª Ordem a Sinais Contínuos

Os Sistemas de 1ª ordem estáveis respondem ao impulso com decaimento exponencial Já os Sistemas de 2ª ordem estáveis podem responder com decaimento exponencial, com oscilação amortecida, ou com uma oscilação pura. Este comportamento depende dos coeficientes da equação diferencial.

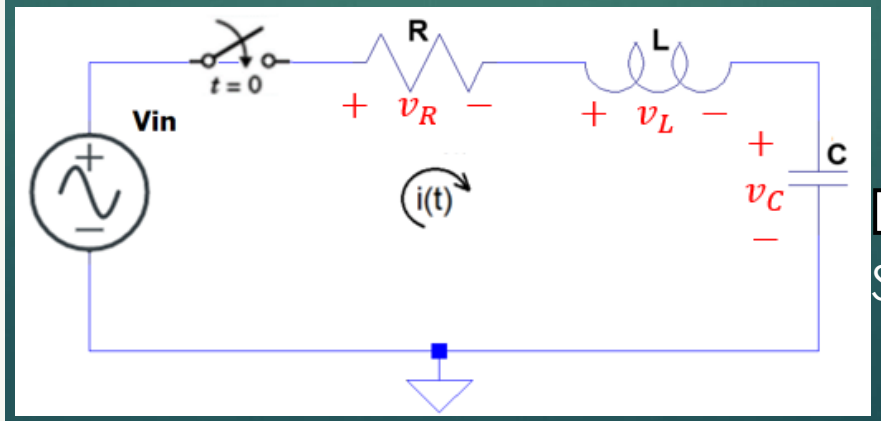
A resposta de sistema de 2ª ordem a um sinal de entrada $x(t)$ também pode ser obtida pela Convolução do sinal de entrada $x(t)$ com a resposta ao impulso do sistema $h(t)$. $y(t) = x(t) * h(t)$

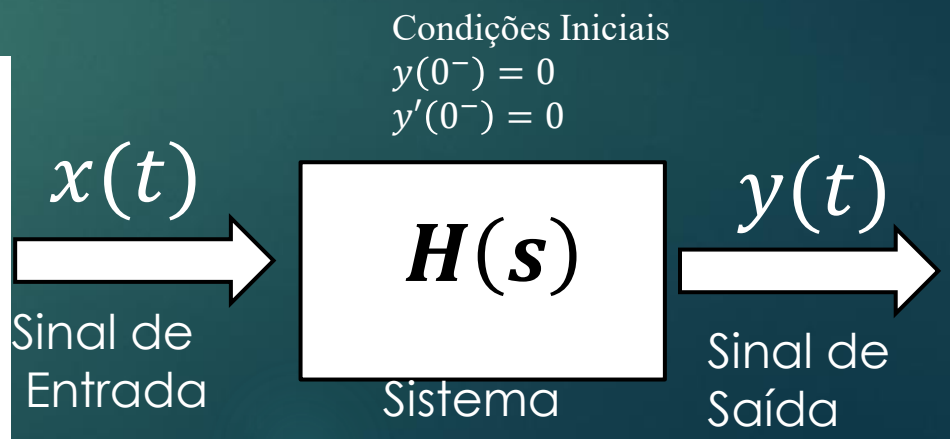
A resposta ao impulso $h(t)$ pode ser obtida diretamente da equação diferencial que define o sistema , ou mesmo ser diretamente medida em um ensaio de laboratório.

Este sistema pode representar o consumo elétrico de uma Residência. R_{eq} = Aquecedores, lâmpadas, ferros de passar roupa, eletrônicos, cargas em geral. L_{eq} = Ar condicionado Central, bomba d'água, pequenos motores. C_{eq} = Sistema de energia Solar, filtros de linha, sistemas de áudio com sub-woofer.

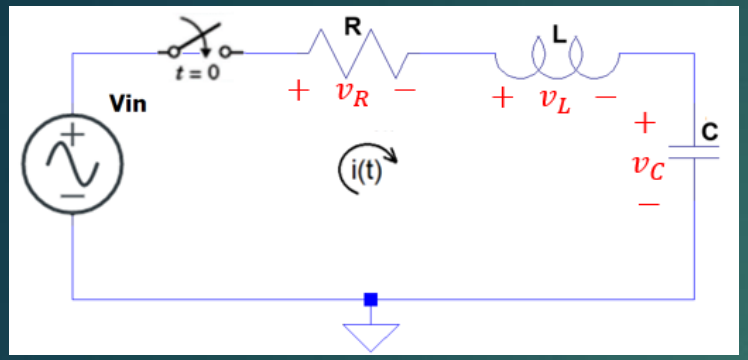


$$R_{eq} = 12\Omega; \quad L_{eq} = 0.1H; \quad C_{eq} = 3.7mF$$





Circuito eletrônico RLC



Condições Iniciais
$v_c(0^-) = 0$
$v_c'(0^-) = 0$

$x(t) = v_{in}(t)$ → $\boxed{H(s)}$ → $y(t) = v_c(t)$

Sinal de Entrada — Sistema — Sinal de Saída

Condições Iniciais
$v_c(0^-) = 0$
$v_c'(0^-) = 0$

$Malha \quad v_R(t)+v_L(t)+v_C(t) = v_{in}(t) \;\; (1)$

$v_R(t) = R\, i(t)$

$v_L(t) = Li'(t)$

$v_{in} = R\, i(t) + Li'(t) + v_C(t) \;\; (1)$

Mas no capacitor $i(t) = C\, v_c'(t)$

$i'(t) = C\, v_c''(t)$

Então :

$R\, Cv_c'(t) + LCv_c''(t) + v_C(t) = \mathrm{v_{in}(t)}$

$$v_c''(t) + \frac{R}{L} v_c'(t) + \frac{1}{LC} v_c(t) = \frac{1}{LC} v_{in}(t)$$

$R = 12\Omega$

$L = 0.1H$

$C = 3.7mF$ teremos:

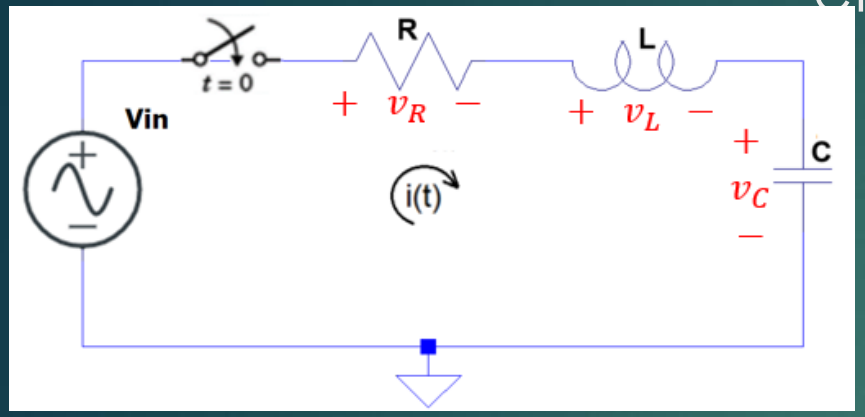
$$v_c''(t) + 120 v_c'(t) + 2700 v_c(t) = 2700\; v_{in}(t)$$

$x(t) = v_{in}(t)$

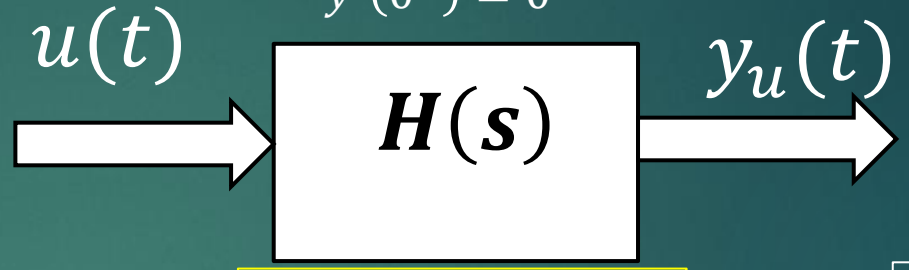
$y(t) = v_c(t)$

$$y''(t) + 120 y'(t) + 2700 y(t) = 2700 x(t)$$

Calcular através da Equação Diferencial a resposta deste sistema para as entradas Degrau e Impulso



$$y(0^-) = 0$$
$$y'(0^-) = 0$$



$$H(s) = \frac{2700}{(s+30)(s+90)}$$

$$y''(t) + 120y'(t) + 2700y(t) = 2700x(t)$$

$x(t) \rightarrow \boldsymbol{X}(s)$

$x'(t) \rightarrow s\boldsymbol{X}(s)$

$y(t) \rightarrow s\boldsymbol{Y}(s)$

$y'(t) \rightarrow sY(s)$

$y''(t) \rightarrow s^2Y(s)$

$$\boldsymbol{H}(s) = \frac{\boldsymbol{Y}(s)}{\boldsymbol{X}(s)}$$

- Calcular primeiro a resposta ao degrau $u(t)$

Fazemos $x(t) = 1\ para\ t \geq 0$ e mantemos as condições iniciais nulas

- $y(0^-) = 0$
- $y'(0^-) = 0$
- A Resposta ao Degrau $y(t)$ será a soma da resposta homogênea mais a resposta forçada
- $y_u(t) = y_h(t) + y_f(t)$
- A Reposta ao Impulso $\mathrm{h(t)} = y_u{}'(t)$ será a derivada da Resposta ao Degrau

$$y''(t) + 120y'(t) + 2700y(t) = 2700x(t)$$
$$s^2Y(s) + 120sY(s) + 2700Y(s) = 2700X(s)$$
$$H(s) = \frac{Y(s)}{X(s)} = \frac{2700}{s^2 + 120s + 2700} = \frac{2700}{(s+30)(s+90)}$$

1) Resposta Homogênea ou Resposta Natural

$$y''(t) + 120y'(t) + 2700y(t) = 0$$

$$\alpha = \frac{120}{2} = 60 \;\; e \;\; w_0^2 = 2700; \;\; w_o = 51.96$$

$$\beta = \sqrt{\alpha^2 - w_0^2} = \sqrt{60^2 - 2700} = \pm 30$$

$$y_h = e^{-\alpha t}\left(A_1 e^{\beta t} + A_2 e^{-\beta t}\right) = e^{-60t}\left(A_1 e^{30t} + A_2 e^{-30t}\right)$$

$$y_h(t) = A_1 e^{-30t} + A_2 e^{-90t}$$

$$\boldsymbol{x(t) = 10e^{-15t}u(t)}$$

Resposta Homogênea 2ª ordem

1) $\alpha > w$ $\quad y_h(t) = A_1 e^{(-\alpha+\beta)t} + A_2 e^{(-\alpha-\beta)t}$
2) $\alpha < w$ $\quad y_h(t) = e^{-\alpha t}[A_1 \text{sen}(\beta t) + A_2 \cos(\beta t)]$
3) $\alpha = w$ $\quad y_h(t) = e^{-\alpha t}[A_1 + A_2 t]$
4) $\alpha = 0$ $\quad y_h(t) = A_1 \text{sen}(\beta t) + A_2 \cos(\beta t)$

Entrada Forçada	Resposta forçada
$f(t) = k_1$	$y_f(t) = B_1$
$f(t) = k_1 t + k_2$	$y_f(t) = B_1 t + B_2$
$f(t) = k_1 sen(w_1 t)$	$y_f(t) = B_1 \cos(w_1 t) + B_2 sen(w_1 t)$
$f(t) = k_1 \cos(w_1 t)$	$y_f(t) = B_1 \cos(w_1 t) + B_2 sen(w_1 t)$
$f(t) = k_1 e^{\beta t}$	$y_f(t) = B_1 e^{\beta t}$
$f(t) = k_1 \cos(w_1 t) + k_2 sen(w_1 t)$	$y_f(t) = B_1 \cos(w_1 t) + B_2 sen(w_1 t)$

2) Resposta Forçada

$$x(t) = 1 \; para \; t \geq 0$$

$$y_f(t) = B_1 e^{-15t}$$

(a resposta forçada terá a mesma forma da entrada)

$$(-15)(-15)B_1 e^{-15t} + 120B_1(-15)e^{-15t} + 2700B_1 e^{-15t} = 2700x($$

$$(10)(0 + 2700B_1 = 2700$$

$$B_1 = 1$$

$$y_f(t) = 1$$

3) Resposta ao Degrau

$$y_u(t) = y_h(t) + y_f(t) \;\; = (A_1 e^{-30t} + A_2 e^{-90t} + 1) \; (1)$$

Agora usamos as condições iniciais :

$$y_u(0) = 0 = A_1 e^{-0} + A_2 e^{-0} + 1$$

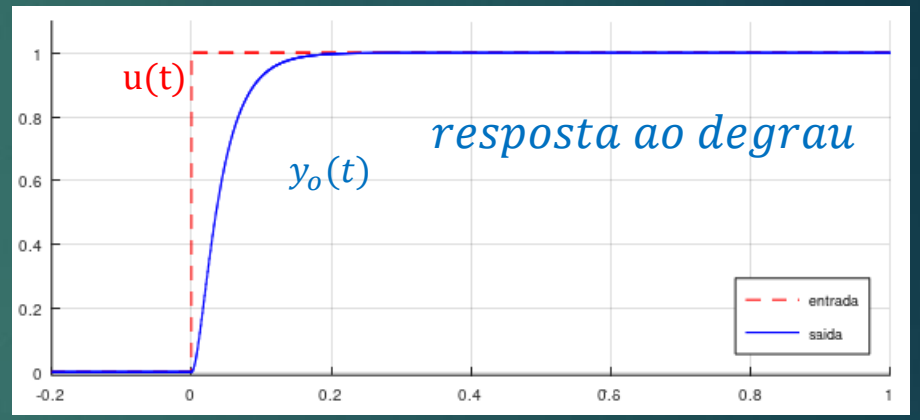
$$y_u'(t) = -30A_1 e^{-30t} - 90A_2 e^{-90t} \; (6)$$

$$y_u'(0) = 0 = -30A_1 e^{-0} - 90A_2 e^{-0}$$

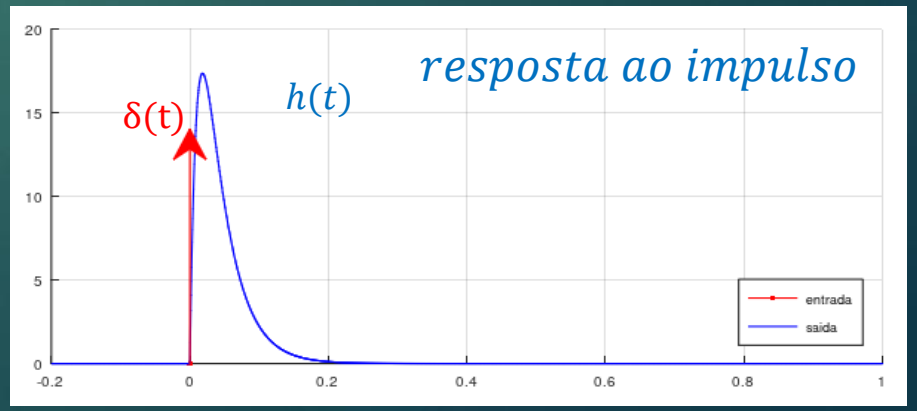
$$\begin{cases} A_1 + A_2 = -1 \\ -30A_1 - 90A_2 = 0 \end{cases}$$

$$A_1 = -1.5 \quad A_2 = 0.5'$$

```
>> exemplo_B1()
```




```
--> exemplo_B2()
```



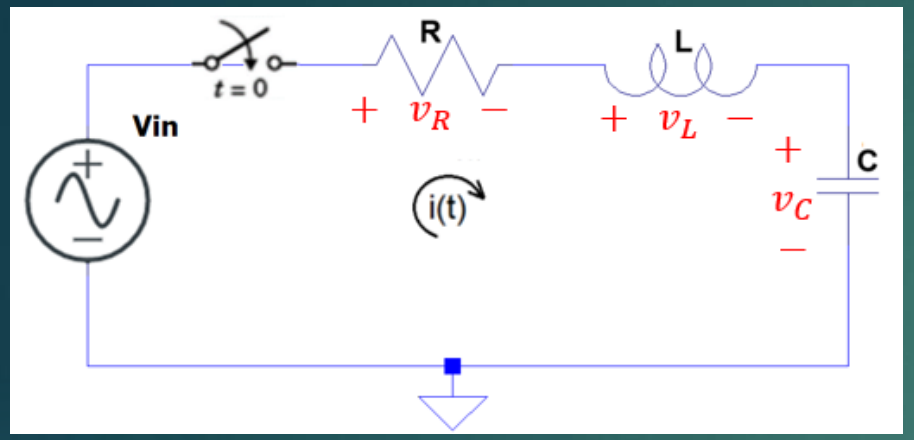
Resposta ao Degrau

$$y_u(t) \;\; = (1 - 1.5e^{-30t} + 0.5e^{-90t})u(t)$$

Resposta ao Impulso é a derivada da resposta ao Degrau

$$h(t) = y'_u(t) \;\; = (45e^{-30t} - 45e^{-90t})u(t)$$

Calcular por **Convolução** a Resposta ao Estado Nulo $y_o(t)$ deste sistema para entrada $x(t) = 10e^{-15t}\mathrm{u(t)}$



$x(t)$ → Sinal de Entrada

$$y_o(t) = x(t) * h(t)$$

$$h(t) = 45e^{-30t} - 45e^{-90t}$$

$y_o(t)$ → Sinal de Saída

$$H(s) = \frac{2700}{(s+30)(s+90)}$$

$$y_o(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

$$y_o(t) = \left((45e^{-30t} - 45e^{-90t})u(t)\right) * \left(10e^{-15t}u(t)\right)$$

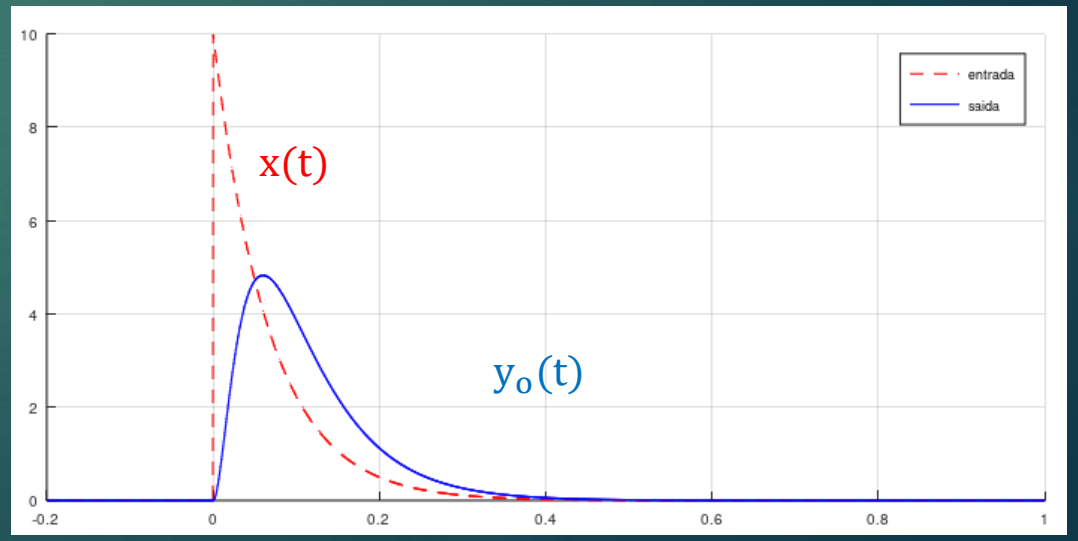
$$y_o(t) = \int_{-\infty}^{\infty} (45\,e^{-30\tau} - 45\,e^{-90\tau})\mathrm{u}(\tau)\left(10e^{-15(t-\tau)}u(t-\tau)\right)d\tau$$

$$y_o(t) = 450\int_{0}^{t} (e^{-30\tau} - e^{-90\tau})e^{-15(t-\tau)}d\tau$$

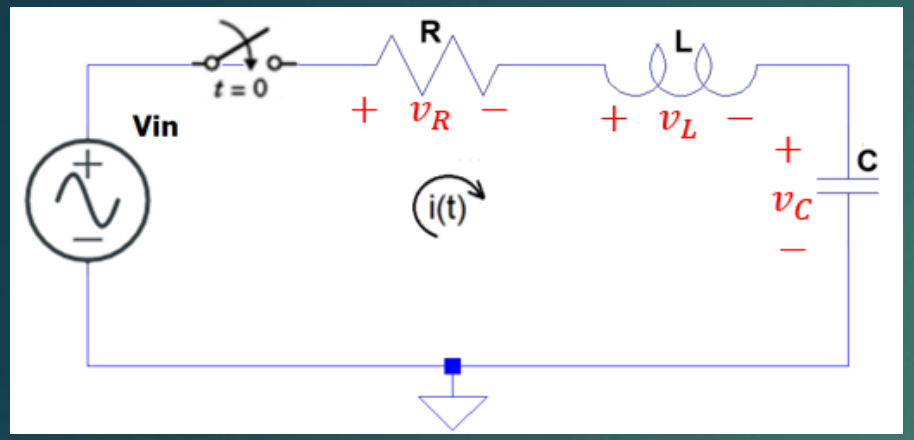
$$y_o(t) = 6(1 - 5e^{60\tau})e^{-15(t+5\tau)}\Big|_{0}^{t} = 6(4e^{-15t} - 5e^{-60t} + e^{-90t})\mathrm{u(t)}$$

$$\boxed{y_o(t) = (24e^{-15t} - 30e^{-30t} + 6e^{-90t})\mathrm{u(t)}}$$

```
>> exemplo_B3o()
```



Calcular por Convolução a resposta ao Estado Nulo $y_o(t)$ deste sistema para entrada $x(t) = 37\cos(15t)u(t)$



$x(t)$ → Sinal de Entrada

$$y_o(t) = x(t) * h(t)$$

$$h(t) = 45e^{-30t} - 45e^{-90t}$$

→ $y_o(t)$ Sinal de Saída

$$H(s) = \frac{2700}{(s+30)(s+90)}$$

$$y_o(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

$$y_o(t) = \left((45e^{-30t} - 45e^{-90t})u(t)\right) * \left(37\cos(60t)\,u(t)\right)$$

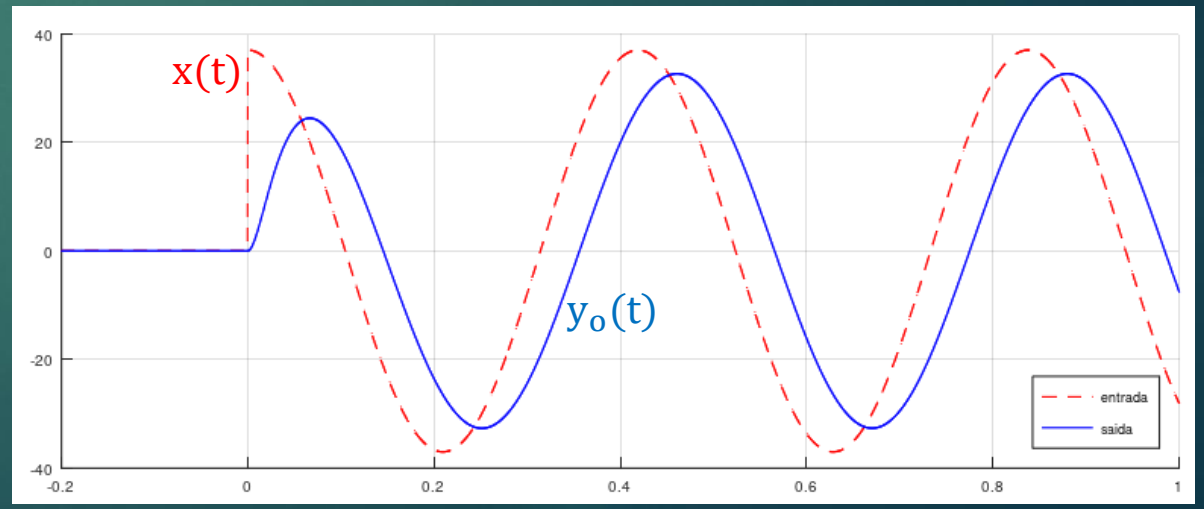
$$y_0(t) = \int_{-\infty}^{\infty} (45\,e^{-30\tau} - 45\,e^{-90\tau})\mathrm{u}(\tau)\left(37\cos\big(15(t-\tau)\big)\,u(t-\tau)\right)d\tau$$

$$y_o(t) = 1665\int_0^t (e^{-30\tau} - e^{-90\tau})\cos\big(15(t-\tau)\big)\,d\tau$$

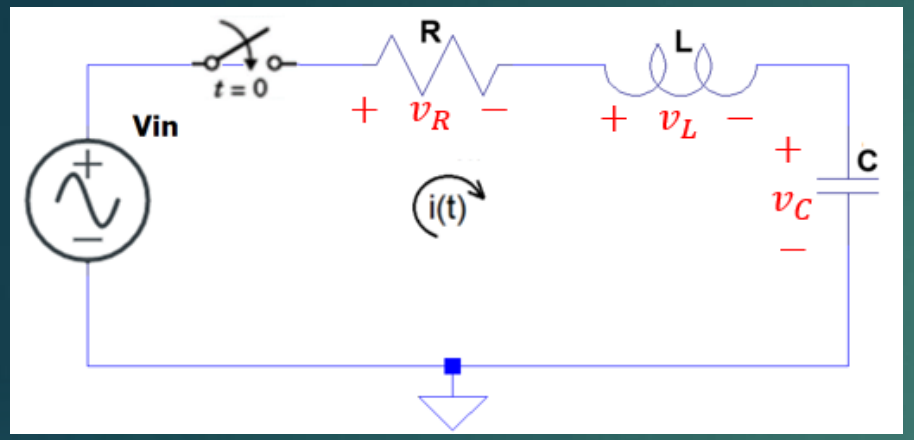
$$y_o(t) = (-222e^{-30\tau} + 90e^{-90\tau})\cos\big(15(t-\tau)\big) + (111e^{-30\tau} - 15e^{-90\tau})\sin\big(15(t-\tau)\big)\Big|_0^t$$

$$y_0(t) = (-44.4e^{-30t} + 18e^{-90t} + 26.4cos(15t) + 19.2\,sin(15t))\mathrm{u}(t)$$

```
>> exemplo_B4o()
```



Calcular a Resposta Completa $y_c(t)$ deste sistema para entrada $x(t) = 37\cos(15t)$, com **condições iniciais** $y(0^-) = 20\text{V}$ e $y'(0^-) = 0$



Condições Iniciais
$y(0^-) = 20$
$y'(0^-) = 0$

$x(t)$ → Sinal de Entrada

$$\boldsymbol{y_o(t) = x(t) * h(t)}$$
$$h(t) = 45e^{-30t} - 45e^{-90t}$$
$$y_c(t) = y_0(t) + y_n(t)$$

→ $y_c(t)$ Sinal de Saída

$$H(s) = \frac{2700}{(s+30)(s+90)}$$

- Calcular $y(t)$ *se* $x(t) = 10\cos(4t)$
- Condições Iniciais
- $y_c(0^-) = 20V$
- $y_c'(0^-) = 0$

Com condições iniciais, precisamos simplesmente somar a **Resposta à Entrada Nula** $y_n(t)$ à **Resposta ao Estado Nulo** $y_0(t)$ que já foi calculada

$$y_c(t) = y_0(t) + y_n(t)$$

$y_n(t)$ é semelhante á reposta homogênea, incluindo as condições iniciais

4) Resposta à Entrada Nula $y_n(t)$ é semelhante á reposta homogênea, incluindo as condições iniciais

$y_n(t) = A_1 e^{-30t} + A_2 e^{-90t}$

Agora usamos as condições iniciais :

$y_n(0) = 20 = A_1 e^{-0} + A_2 e^{-0} = A_1 + A_2$

$y_n'(t) = -30A_1 e^{-30t} - 90A_2 e^{-90t}$ (5)

$y_n'(0) = 0 = -30A_1 e^{-0} - 90A_2 e^{0}$

$$\begin{cases} A_1 + A_2 = 20 \\ 30 + 90A_2 = 0 \end{cases}$$

$A_1 = -30 \;\; ; \;\; A_2 = 10$

$y_n(t) = 30e^{-30t} - 10e^{-90t}$

```
>> exemplo_B4o()
```

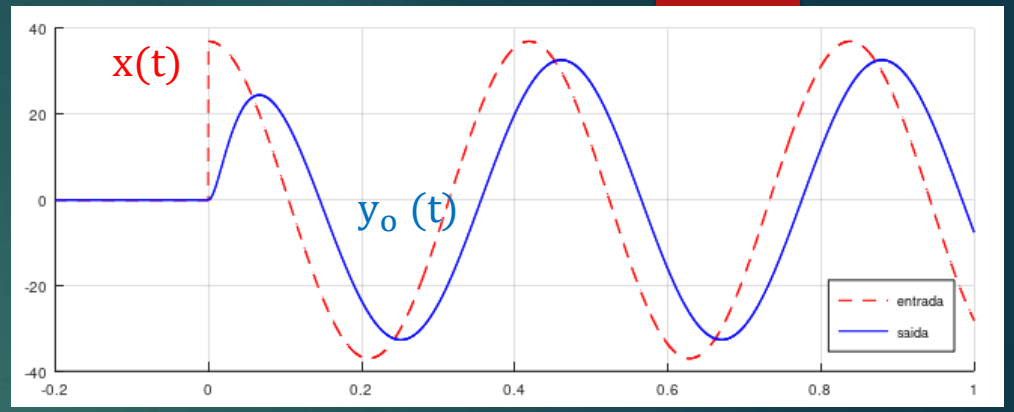
Resposta Homogênea 2ª ordem

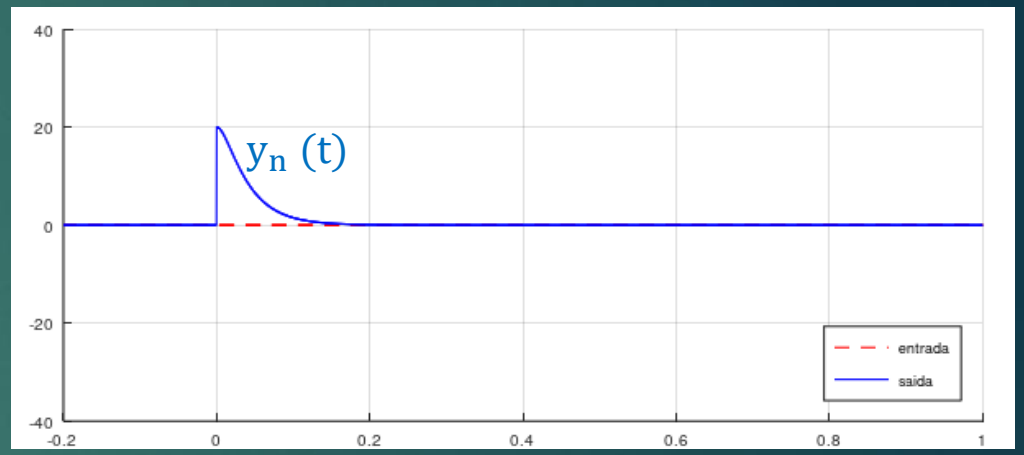
1) $\alpha > w$ $\quad y_h(t) = A_1 e^{(-\alpha+\beta)t} + A_2 e^{(-\alpha-\beta)t}$

2) $\alpha < w$ $\quad y_h(t) = e^{-\alpha t}[A_1 \mathrm{sen}(\beta t) + A_2 \cos(\beta t)]$

3) $\alpha = w$ $\quad y_h(t) = e^{-\alpha t}[A_1 + A_2 t]$

4) $\alpha = 0$ $\quad y_h(t) = A_1 \mathrm{sen}(\beta t) + A_2 \cos(\beta t)$





5) Solução Completa será a soma da Resposta ao Estado Nulo $y_0(t)$ com a resposta à Entrada Nula $y_n(t)$

```
>> exemplo_B4n()
```

$y_c(t) = y_{o(t)} + y_n(t)$

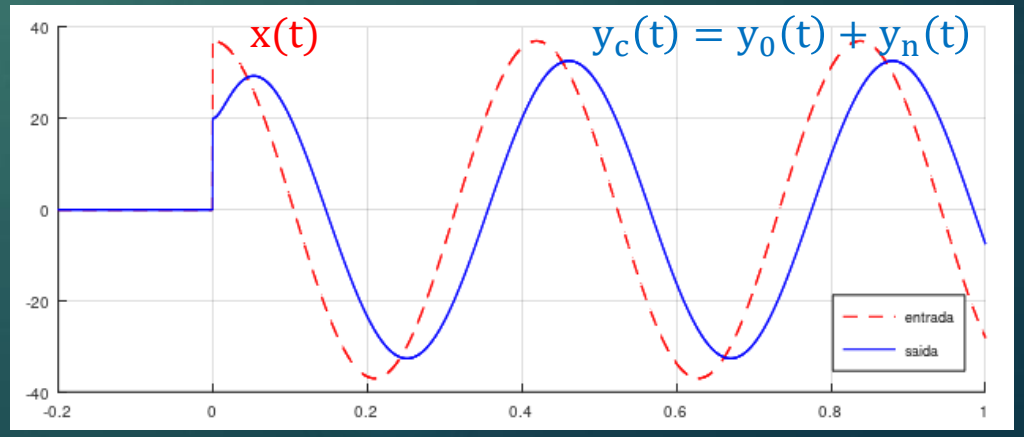
$y_c(t) = (-44.4e^{-30t} + 18e^{-90t} + 26.4cos(15t) + 19.2\, sin(15t)) + (30e^{-30t} - 10e^{-90t})$

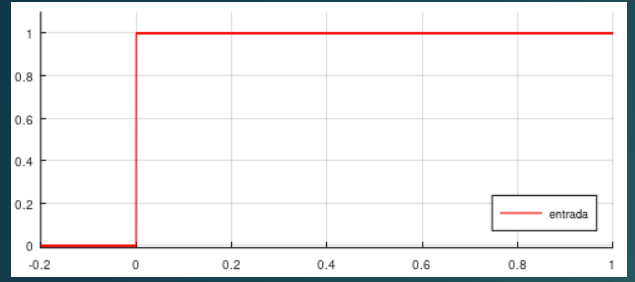
$y_c(t) = -14.4e^{-30t} + 8e^{-90t} + 26.4cos(15t) + 19.2\, sin(15t)$

```
>> exemplo_B4c()
```

$y_c(t) = -14.4e^{-30t} + 8e^{-90t} + \sqrt{26.4^2 + 19.2^2}\, cos(15t - atan2(19.2,26.4)\,)$

$y_c(t) = -14.4e^{-30t} + 8e^{-90t} + 32.6\, cos(15t - 0.6288\,)$



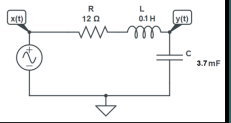


$x(t) = u(t)$

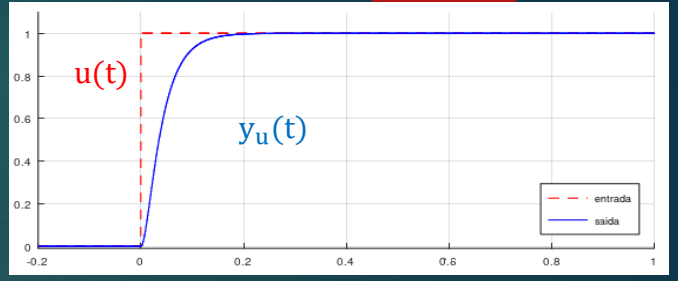
$y''(t) + 120y'(t) + 2700y(t) = 2700x(t)$

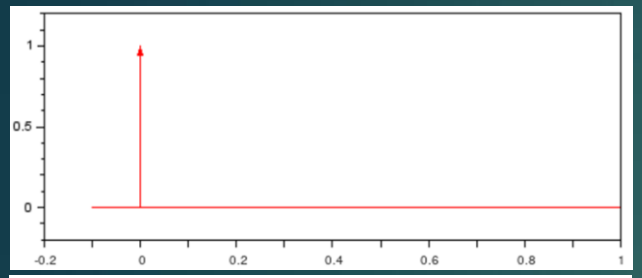
$y(0^-) = 0$

$y'(0^-) = 0$



$y_u(t) = 1 - 1.5e^{-30t} + 0.5e^{-90t}$





$x(t) = \delta(t)$

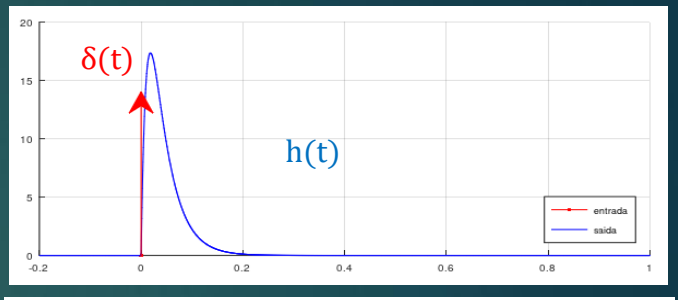
$y''(t) + 120y'(t) + 2700y(t) = 2700x(t)$

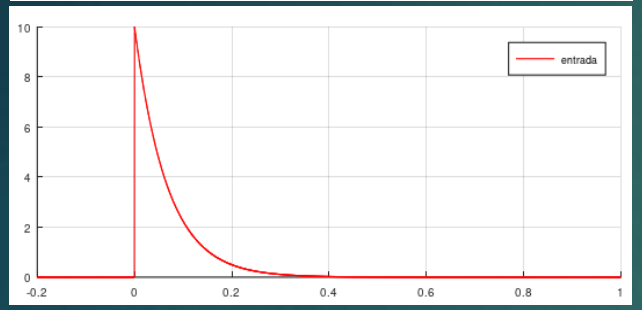
$h(t) = y_u{}'(t)$

$y(0^-) = 0$

$y'(0^-) = 0$

$h(t) = 45e^{-30t} - 45e^{-90t}$



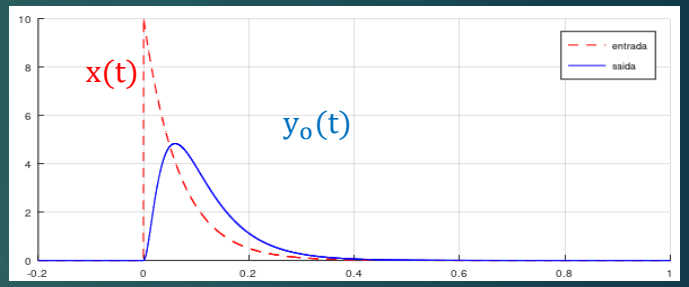


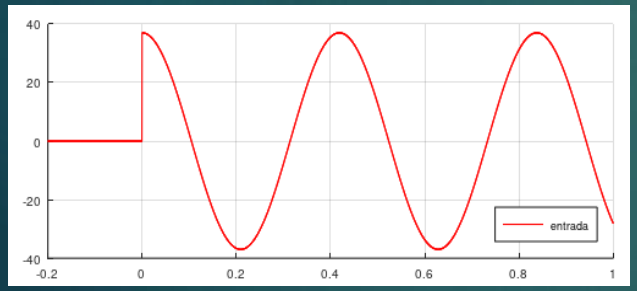
$x(t) = 10e^{-t}$

$h(t) = 45e^{-30t} - 45e^{-90t}$

$y_o(t) = x(t) * h(t)$

$y_o(t) = 24e^{-15t} - 30e^{-30t} + 6e^{-90t}$



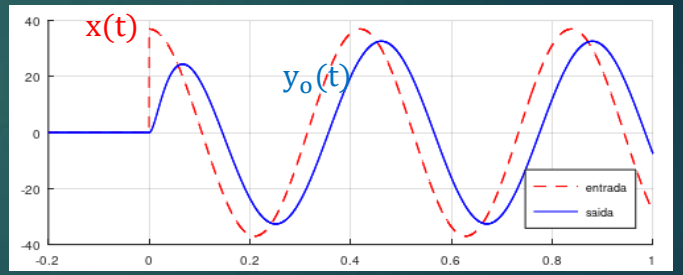


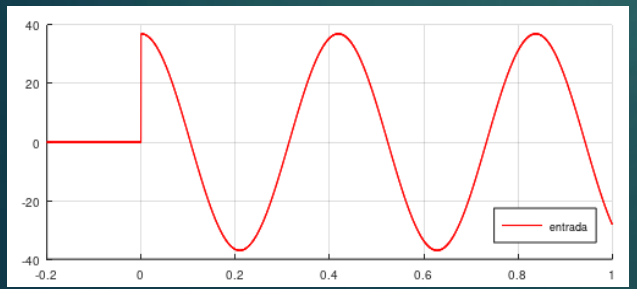
$x(t) = 37\cos(15t)$

$h(t) = 45e^{-30t} - 45e^{-90t}$

$y_o(t) = x(t) * h(t)$

$y_o(t) = -44.4e^{-30t} + 18e^{-90t} + 26.4cos(15t) + 19.2\,sin(15t)$





$x(t) = 37\cos(15t)$

$h(t) = 45e^{-30t} - 45e^{-90t}$

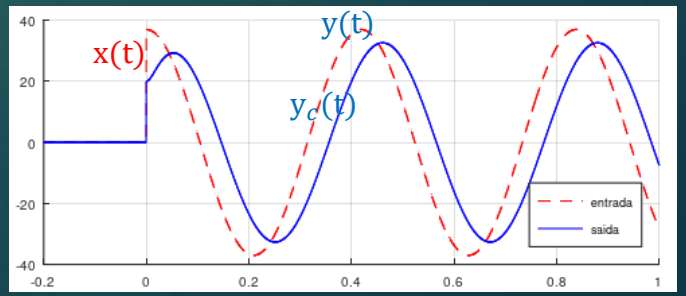
$y_o(t) = x(t) * h(t)$

$y(0^-) = 20$

$y'(0^-) = 0$

$y_c(t) = y_{o(t)} + y_n(t)$

$y_c(t) = -14.4e^{-30t} + 8e^{-90t} + 26.4cos(15t) + 19.2\,sin(15t)$



Equação de Diferenças a partir da Equação Diferencial

Transformar um sistema analógico em digital

$$y''(t) + 120y'(t) + 2700y(t) = 2700x(t)$$

$$y(t) \cong y[n]$$
$$y'(t) \cong \frac{y[n] - y[n-1]}{T}$$
$$y''(t) \cong \frac{y[n] - 2y[n-1] + y[n-2]}{T^2}$$
$$x(t) \cong x[n]$$

$$h(t) = 45e^{-30t} - 45e^{-90t}$$

$$ct = \frac{1}{30}s$$

$$T = \frac{ct}{5} = \frac{\left(\frac{1}{30}\right)}{5} = \frac{1}{150}$$

Diferença Finitas

$$\frac{y[n] - 2y[n-1] + y[n-2]}{T^2} + 120\frac{y[n] - y[n-1]}{T} + 2700y[n] = 2700x[n]$$

$$y[n] - 2y[n-1] + y[n-2] + 120T\,y[n] - 120\,Ty[n-1] + 2700\,T^2\,y[n] = 2700\,T^2\,x[n]$$

$$(1 + 120T + 2700T^2)y[n] + (-2 - 120T)y[n-1] + y[n-2] = 2700\,T^2\,x[n]$$

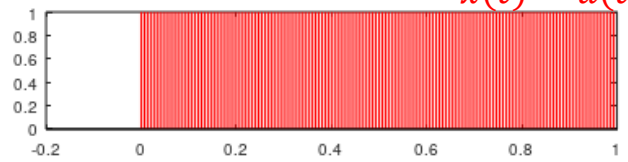
$$y[n] = \frac{(2 + 120T)}{(1 + 120T + 2700T^2)}y[n-1] + \frac{-1}{(1 + 120T + 2700T^2)}y[n-2] + \frac{2700\,T^2}{(1 + 120T + 2700T^2)}x[n]$$

$$T = \frac{1}{150}$$

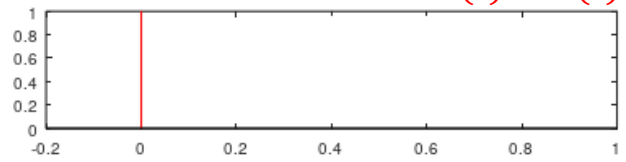
$$y[n] = \frac{35}{24}y[n-1] - \frac{25}{48}y[n-2] + \frac{1}{16}x[n]$$

```
function y_out=Sistema_Eletrico_2aOrdem(x_in,y1,y2,T)
    den=1+120*T+2700*T^2;
    a1=(2+120*T)/den;
    a2=-1/den;
    b0=(2700*T^2)/den;
    xi=[0 0 x_in];
    yi=[y1 y2];
    for (i=3:length(xi))
        yi(i)=a1*yi(i-1)+a2*yi(i-2)+b0*xi(i);
    end
    y_out=yi(3:length(xi));
endfunction
```

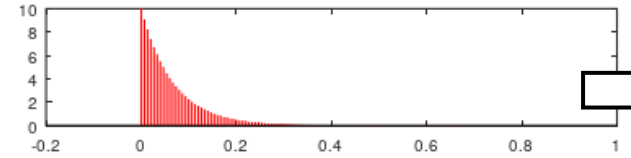
$x(t) = u(t)$



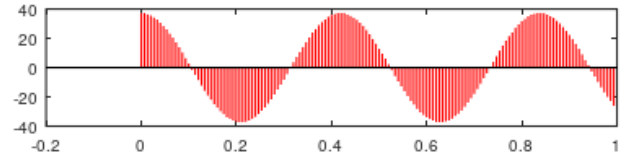
$x(t) = \delta(t)$



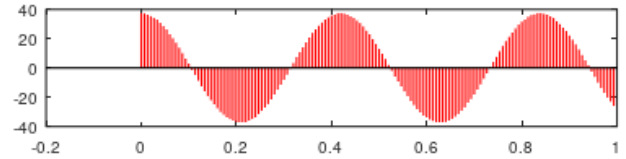
$x(t) = 10e^{-t}$



$x(t) = 10\cos(4t)$



$x(t) = 10\cos(4t)$



$$y[n] = \frac{(2+5T)}{(1+5T+6T^2)}y[n-1] + \frac{-1}{(1+5T+6T^2)}y[n-2] + \frac{6\,T^2}{(1+5T+6T^2)}x[n]$$

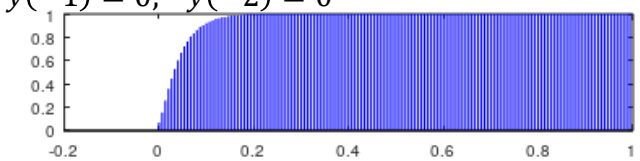
```
function y_out=Sistema_Eletrico_2aOrdem(x_in,y1,y2,T)
    den=1+120*T+2700*T^2;
    a1=(2+120*T)/den;
    a2=-1/den;
    b0=(2700*T^2)/den;
    xi=[0 0 x_in];
    yi=[y1 y2];
    for (i=3:length(xi))
        yi(i)=a1*yi(i-1)+a2*yi(i-2)+b0*xi(i);
    end
    y_out=yi(3:length(xi));
endfunction
```

$$y[n] = \frac{35}{24}y[n-1] - \frac{25}{48}y[n-2] + \frac{1}{16}x[n]$$

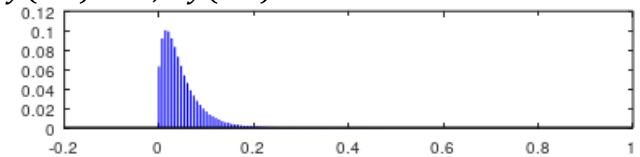
$$T = \frac{1}{150}\,\mathrm{s}$$

```
>> exemplo_B5(1/75)
```

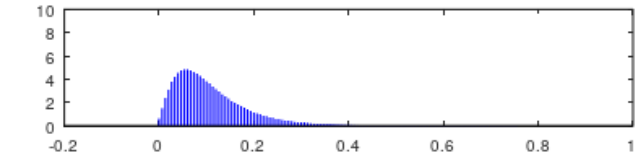
$y(-1) = 0;\quad y(-2) = 0$



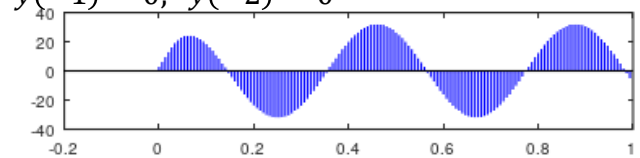
$y(-1) = 0;\quad y(-2) = 0$



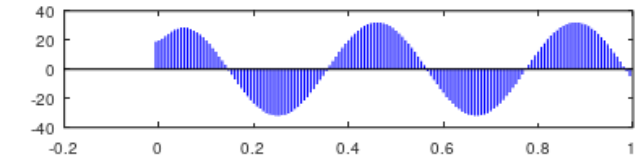
$y(-1) = 0;\quad y(-2) = 0$



$y(-1) = 0;\quad y(-2) = 0$

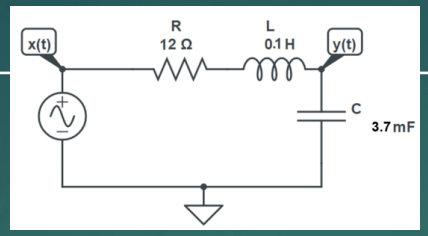


$y(-1) = 20;\quad y(-2) = 20$



a) Analisamos a equação diferencial e encontramos a sua resposta ao Degrau $y_u(t)$ e depois sua Resposta ao Impulso $h(t)$

$$y''(t) + 120y'(t) + 2700y(t) = 2700x(t)$$



1) $y_o(t) = y_h(t) + y_f(t)$

1.1) $y_h(t) = A_1e^{-30t} + A_2e^{-90t}$

A_1 e A_1 são avaliadas em (1.3)

1.2) $y_f(t) = B$

Encontrar B substituindo $y_f(t)$ na eq. diferencial

1.3) $y_u(t) = (A_1e^{-30t} + A_2e^{-90t}) + B = 1 - 1.5e^{-30t} + 0.5e^{-90t}$

Encontrar A_1 e A_1 substituindo as condições nulas em $y_o(t)$

1.4) $h(t) = y'_u(t) = 45e^{-30t} - 45e^{-90t}$

Resposta Homogênea 2ª ordem

1) $\alpha > w$ $\quad y_h(t) = A_1e^{(-\alpha+\beta)t} + A_2e^{(-\alpha-\beta)t}$

2) $\alpha < w$ $\quad y_h(t) = e^{-\alpha t}[A_1\text{sen}(\beta t) + A_2\cos(\beta t)]$

3) $\alpha = w$ $\quad y_h(t) = e^{-\alpha t}[A_1 + A_2 t]$

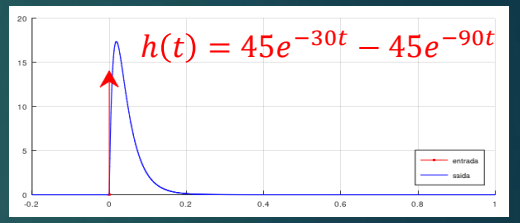
4) $\alpha = 0$ $\quad y_h(t) = A_1\text{sen}(\beta t) + A_2\cos(\beta t)$

Resposta Forçada

$x(t) = 1$ $\qquad y_f(t) = B$

b) Encontramos a Resposta ao Estado Nulo $y_o(t)$ a uma entrada genérica $x(t)$ por convolução

$$x(t) \Rightarrow \boxed{H(s) = \frac{2700}{(s+30)(s+90)}} \Rightarrow y_o(t) = x(t) * h(t) = x(t) * (45e^{-30t} - 45e^{-90t})$$



c) Se houver condições iniciais, somamos a resposta à Entrada Nula $y_n(t)$ à resposta ao Estado Nulo $y_o(t)$

$y_n(t)$ tem forma semelhante à resposta homogênea

$y_n(t) = A_1e^{-30t} + A_2e^{-90t}$ Encontrar A_1 e A_2 substituindo as condições iniciais

$y_c(t) = y_0(t) + y_n(t)$ A Resposta completa

d) Encontramos a saída completa discreta $y(n)$ transformado a equação diferencial em uma equação de diferenças (algoritmo). Para isso tivemos que escolher um período de amostragem T, para transforma a função contínua em discreta.

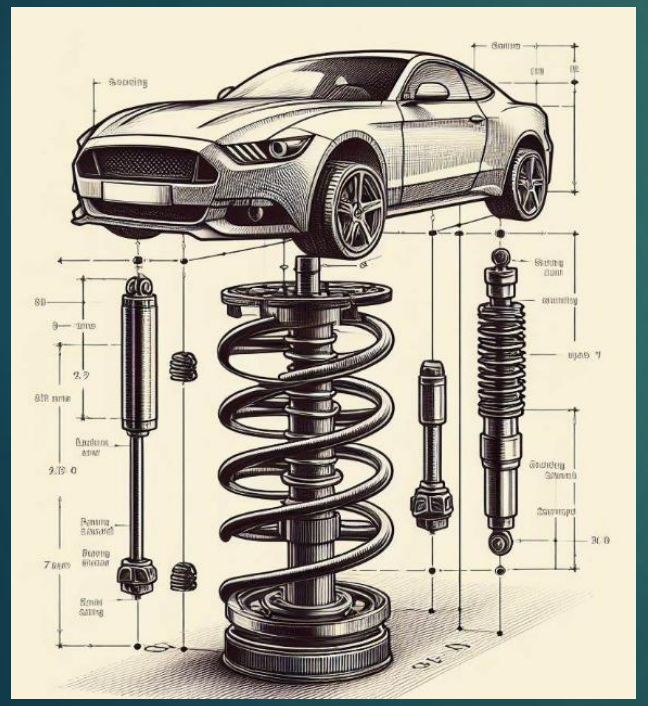
$$y''(t) + 120y'(t) + 2700y(t) = 2700x(t) \xrightarrow{T = 1/150} y[n] = \frac{35}{24}y[n-1] - \frac{25}{48}y[n-2] + \frac{1}{16}x[n]$$

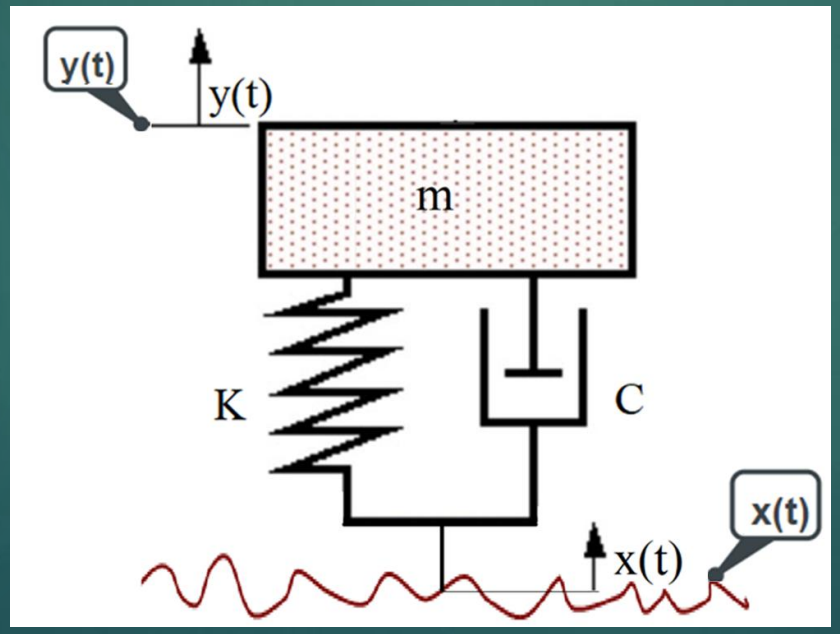
$$H(s) = \frac{2700}{(s+30)(s+90)}$$

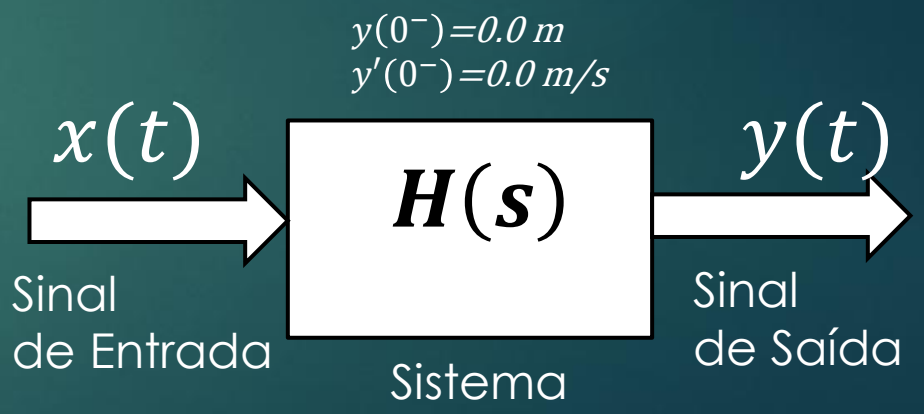
$$H(z) = \frac{(1/16)}{1 - \frac{35}{24}z^{-1} + \frac{25}{48}z \pm 2}$$

C) Resposta de Sistemas Mecânicos CKM de 2ª Ordem a sinais contínuos

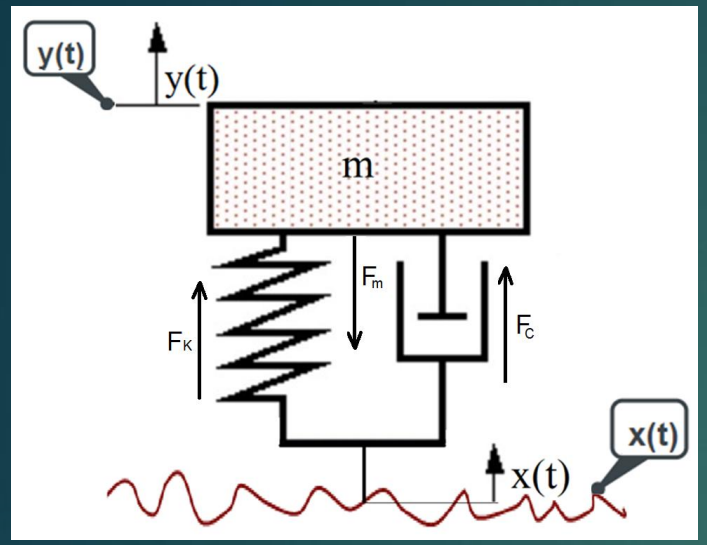
Nos sistemas mecânico temos a massa (m), a mola (K) e o amortecedor(C). O corpo, neste caso o automóvel, é submetido a um perfil de estrada x(t) e responde com a mudança de posição vertical y(t). As condições iniciais são a posição vertical inicial, $y(0^-)$, e a velocidade vertical inicial vertical do automóvel, $y'(0^-)$.

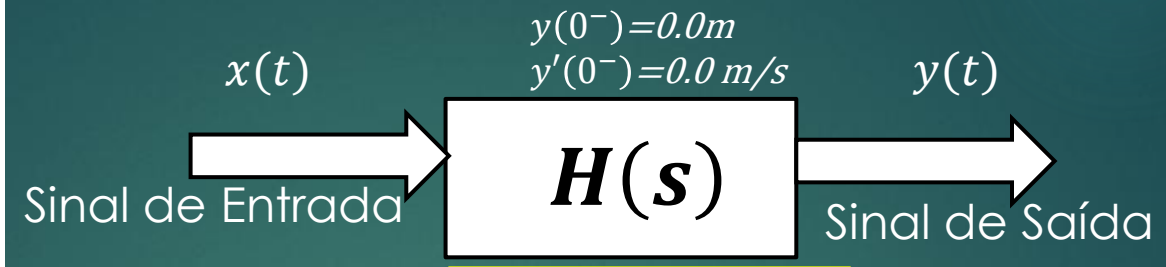






A equação diferencial é obtida pelo equilíbrio de forças $\sum f = f_m(t) + f_c(t) + f_K(t) = 0$





$$H(s) = \frac{4s + 40}{(s+2)^2 + 6^2}$$

- x(t) será o perfil da estrada
- y(t) será o movimento vertical resultante do carro

$$\boldsymbol{my''(t) + C(y'(t) - x'(t)) + k(y(t) - x(t)) = 0}$$

$$\boldsymbol{my''(t) + Cy'(t) + Ky(t) = Cx'(t) + Kx(t)}$$

$$\boldsymbol{y''(t) + \frac{C}{m}y'(t) + \frac{K}{m}y(t) = \frac{C}{m}x'(t) + \frac{K}{m}x(t)}$$

$$\sum f = f_m(t) + f_c(t) + f_K(t) = 0 \quad (1)$$

- *A força resultante é igual à massa do corpo vezes a sua aceleração*
- $f_m = ma = my''$ (2)
- *A força no amortecedor é igual ao coeficiente de amortecimento viscoso vezes a velocidade do corpo*
- $f_C = Cv = C\,(y' - x')$ (3)
- *A força na mola é igual à rigidez K da mola vezes a posição do corpo.*
- $f_K = kx = K(y - x)$ (4)

m=300kg	*Massa do automóvel*
K= 12000 N/m	*Rigidez da mola*
C=1200 N/(m/s)	*Coeficiente de amortecimento*

$$y''(t) + 4y'(t) + 40y(t) = 4x'(t) + 40x(t)$$

$$y''(t) + 4y'(t) + 40y(t) = 4x'(t) + 40x(t)$$

$$s^2Y(s) + 4sY(s) + 40Y(s) = 4sX(s) + 40X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{4s + 40}{s^2 + 4s + 40} = \frac{4s + 40}{(s+2)^2 + 6^2}$$

$x(t) \rightarrow \boldsymbol{X}(s)$

$x'(t) \rightarrow s\boldsymbol{X}(s)$

$y(t) \rightarrow s\boldsymbol{Y}(s)$

$y'(t) \rightarrow sY(s)$

$y''(t) \rightarrow s^2Y(s)$

$$\boldsymbol{H}(s) = \frac{\boldsymbol{Y}(s)}{\boldsymbol{X}(s)}$$

Calcular através da Equação Diferencial a resposta deste Sistema às entradas Degrau e Impulso

$y''(t) + 4y'(t) + 40y(t) = 4x'(t) + 40x(t)$
$x(t) = u(t)$
$y''(t) + 4y'(t) + 40y(t) = 4\delta(t) + 40u(t)$ (1)
Como na equação diferencial temos o termo em $x'(t)$, o degrau na entrada criará um descontinuidade na derivada no instante de tempo t=0 ($\mathbf{4\delta(t)}$)

$$\boxed{\begin{aligned} y(0^-) &= 0 \\ y'(0^-) &= 0 \end{aligned}} \Rightarrow \boxed{\begin{aligned} y(0^+) &= 0 \\ y'(0^+) &= 4 \end{aligned}}$$

- Calcular primeiro a resposta ao degrau $u(t)$

Fazemos $x(t) = 1 \;\; t \geq 0$ e mantemos as condições iniciais nulas (Atualizadas!!)

- $y(0^-) = 0$
- $y'(0^-) = 4$
- A Resposta ao Degrau $y_u(t)$ será a soma da resposta homogênea mais a resposta forçada
- $y_u(t) = y_h(t) + y_f(t)$
- A Reposta ao Impulso $\mathrm{h(t)} = y_u{}'(t)$ será a derivada da Resposta ao Degrau

1) Resposta Homogênea ou Resposta Natural (Regime transitório)

$y'' + 4y' + 40y = 0$

$\alpha=2$ e $w_0^2=40$ (Equação padrão: $y''(t)+2\alpha\, y'(t) + w_0^2\, y(t)= 0$)

$$\beta = \sqrt{\alpha^2 - w_0^2} = \sqrt{2^2 - 40} = \pm j6$$

$$y_h(t) = e^{-\alpha t}(A_1\cos(\beta t) + A_2 sen(\beta t)) = e^{-2t}(A_1\cos(6t) + A_2 sen(6t))$$

As constante A_1 e A_2 serão avaliadas no final, após incluirmos a resposta forçada

Resposta Homogênea 2ª ordem

1) $\alpha > w$ $\quad y_h(t) = A_1 e^{(-\alpha+\beta)t} + A_2 e^{(-\alpha-\beta)t}$
2) $\alpha < w$ $\quad y_h(t) = e^{-\alpha t}[A_1 sen(\beta t) + A_2\cos(\beta t)]$
3) $\alpha = w$ $\quad y_h(t) = e^{-\alpha t}[A_1 + A_2 t]$
4) $\alpha = 0$ $\quad y_h(t) = A_1 sen(\beta t) + A_2\cos(\beta t)$

2) Resposta Forçada - Regime permanente

$x(t) = 1 \;\; para\; t \geq 0$

$y_f(t) = B_1$ (resposta forçada terá a mesma forma da entrada)

$y''(t) + 6y'(t) + 40y(t) = 6x'(t) + 40x(t)$

$(0) + 6(0) + 40(B_1) = 6\,(0) + 40$

$B_1 = 1$

$y_f(t) = 1$

Entrada Forçada	Resposta forçada
$f(t) = k_1$	$y_f(t) = B_1$
$f(t) = k_1 t + k_2$	$y_f(t) = B_1 t + B_2$
$f(t) = k_1 sen(w_1 t)$	$y_f(t) = B_1\cos(w_1 t) + B_2 sen(w_1 t)$
$f(t) = k_1\cos(w_1 t)$	$y_f(t) = B_1\cos(w_1 t) + B_2 sen(w_1 t)$
$f(t) = k_1 e^{\beta t}$	$y_f(t) = B_1 e^{\beta t}$
$f(t) = k_1\cos(w_1 t) + k_2 sen(w_1 t)$	$y_f(t) = B_1\cos(w_1 t) + B_2 sen(w_1 t)$

3) Resposta ao Degrau

$y_u(t) = y_h(t) + y_f(t) \;\; = e^{-2t}(A_1\cos(6t) + A_2 sen(6t)) + 1$

```
--> exemplo_C1()
```

Agora usamos as novas condições iniciais :

$y_u(0) = e^{-0}(A_1\cos(0) + A_2 sen(0)) + 1$

$A_1 = \;\; -1$

$y_u'(t) = -2e^{-2t}(A_1\cos(6t) + A_2 sen(6t)) + e^{-2t}(-6A_1\sin(6t) + 6A_2 cos(6t))$

$y_u'(0) = 4 = -2e^{0}(A_1\cos(0) + A_2 sen(0)) + e^{-2t}(-6A_1\sin(0) + 6A_2 cos(0))$

$-2A_1 + 6A_2 = 4$

$A_2 = 1/3$

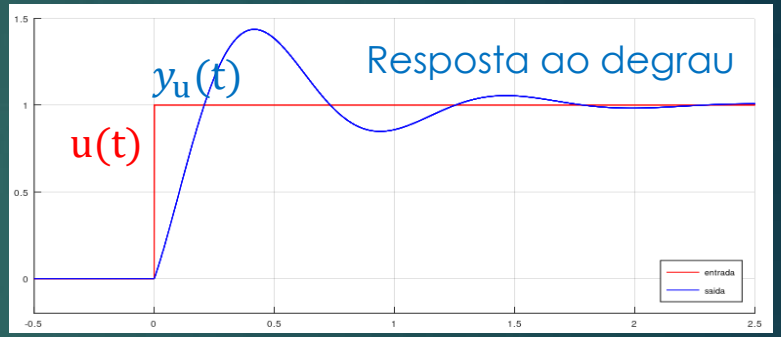
Resposta ao Degrau

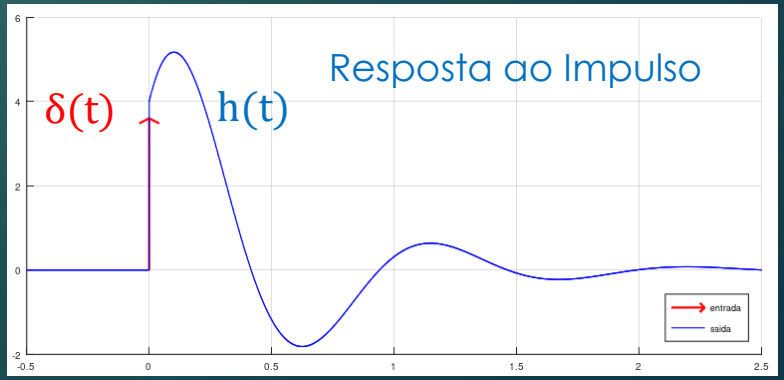
$$y_u(t) = 1 - e^{-2t}\left(\cos(6t) - \frac{1}{3} sen(6t)\right)$$

```
--> exemplo_C2()
```

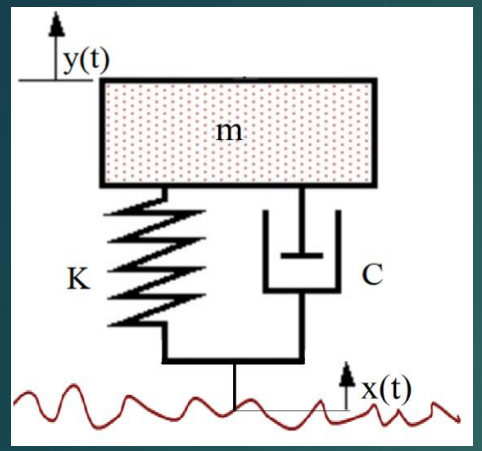
Resposta ao impulso

$$h(t) = y_u'(t) = e^{-2t}\left(4\cos(6t) + \frac{16}{3} sin(6t)\right)$$





Calcular por Convolução a saída ao estado nulo $y_o(t)$ deste sistema para entrada $x(t) = \frac{t}{3}u(t)$



$x(t)$

Sinal de Entrada

$$h(t) = e^{-2t}\left(4\cos(6t) + \frac{16}{3}sin(6t)\right)$$

$$y_0(t) = h(t) * x(t)$$

$y_o(t)$

Sinal de Saída

$$H(s) = \frac{4s + 40}{(s+2)^2 + 6^2}$$

$$y_0(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

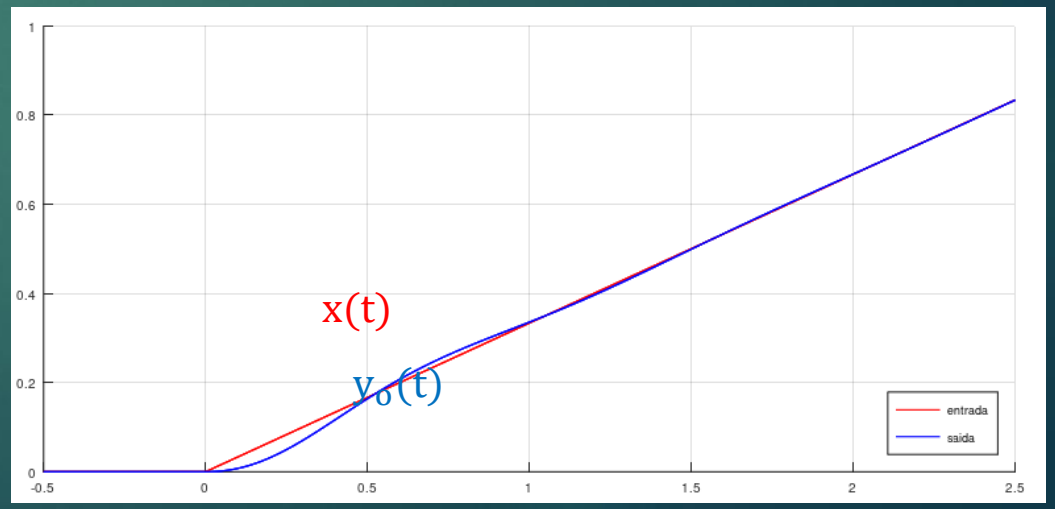
$$y_o(t) = \int_{-\infty}^{\infty}\left(e^{-2\tau}\left(4\cos(6\tau) + \frac{16}{3}sen(6\tau)\right)\right)u(\tau)\frac{(t-\tau)}{3}u(t-\tau)d\tau$$

$$y_o(t) = \int_{0}^{t} e^{-2\tau}\left(4\cos(6\tau) + \frac{16}{3}sen(6\tau)\right)\frac{(t-\tau)}{3}d\tau$$

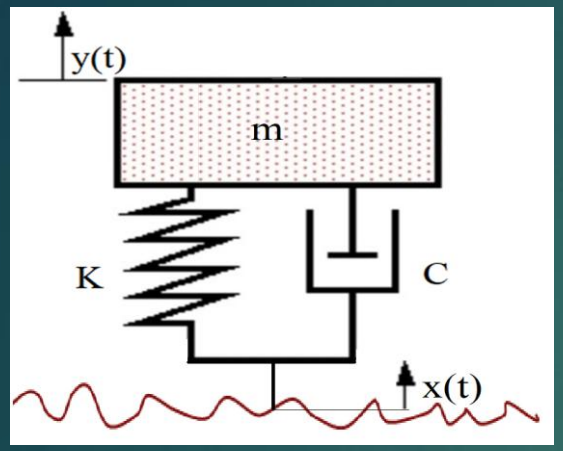
$$y_o(t) = \frac{e^{-2\tau}}{18}\left[(2t - 2\tau - 1)\sin(6\tau) + (6t - 6\tau)\cos(6\tau)\right]\Big|_0^t$$

$$y_0(t) = \left[\frac{1}{3}t - \frac{1}{18}e^{-2t}sen(6t)\right]u(t)$$

```
--> exemplo_C3()
```



Calcular por Convolução a resposta ao Estado Nulo $y_o(t)$ deste sistema para entrada $x(t) = 0.377sen(30t)$



$x(t)$

Sinal de Entrada

$$h(t) = e^{-2t}\left(4\cos(6t) + \frac{16}{3}sin(6t)\right)$$

$$y_0(t) = h(t) * x(t)$$

$y(t)$

Sinal de Saída

$$H(s) = \frac{4s + 40}{(s+2)^2 + 6^2}$$

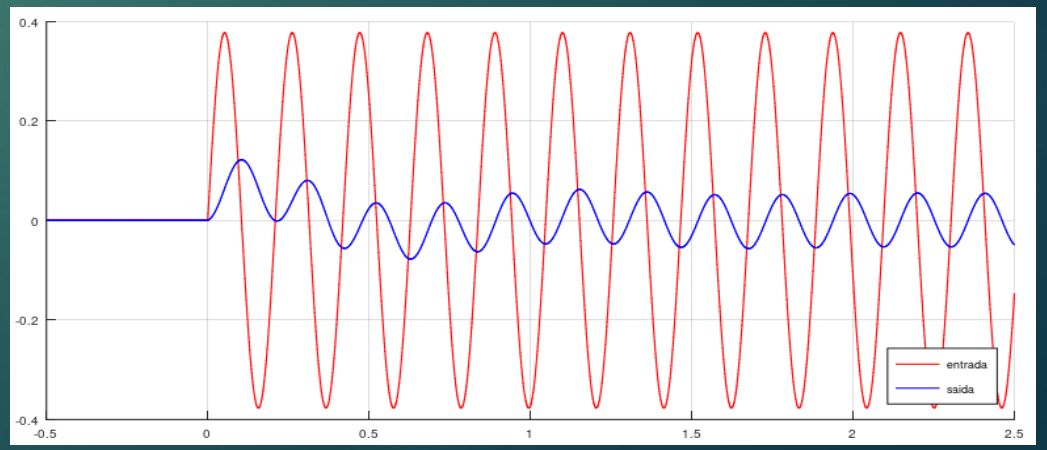
$$y_0(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

$$y_o(t) = \int_{-\infty}^{\infty} \left(e^{-2\tau}\left(4\cos(6\tau) + \frac{16}{3}sen(6\tau)\right)\right) u(\tau) 0.377 \sin\big(30(t-\tau)\big)\, u(t-\tau)d\tau$$

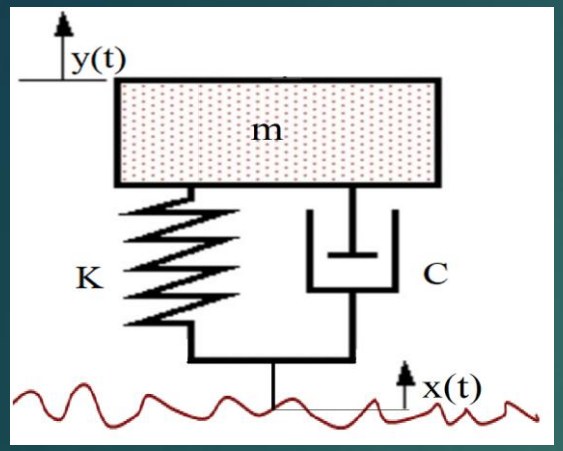
$$y_o(t) = 1.508 \int_0^t e^{-2\tau}\cos(6\tau)\sin\big(30(t-\tau)\big)\,d\tau \; + \frac{6.032}{3}\int_0^t e^{-2\tau}\sin(6\tau)\sin\big(30(t-\tau)\big)\,d\tau$$

$$y_o(t) = e^{-2t}(0.054\cos(6t) + 0.068sen(6t)) - 0.054\cos(30t) - 0.001sen(30t)$$

```
--> exemplo_C4o()
```



Calcular a Resposta Completa $y_c(t)$ deste sistema à entrada $x(t) = 0.377sen(30t)$, com **condições iniciais** $y(0^-) = 0.3$e $y'(0^-) = 0$



condições Iniciais
$y_c(0^-) = 0.3$
$y_c'(0^-) = 0$

$x(t)$ → Sinal de Entrada

$$h(t) = e^{-2t}\left(4\cos(6t) + \frac{16}{3}sin(6t)\right)$$

$$y_0(t) = h(t) * x(t)$$

$$y_c(t) = y_0(t) + y_n(t)$$

→ $y(t)$ Sinal de Saída

$$H(s) = \frac{4s + 40}{(s+2)^2 + 6^2}$$

- Calcular $y(t)$ se $x(t) = 0.377sen(30t)$,
- condições Iniciais
- $y_c(0^-) = 0.3$
- $y_c'(0^-) = 0$

Com condições iniciais, precisamos simplesmente somar a **Resposta à Entrada Nula** $y_n(t)$ à **Resposta ao Estado Nulo** $y_0(t)$ que já foi calculada pela convolução.

$y_c(t) = y_0(t) + y_n(t)$

$y_n(t)$ é semelhante á reposta homogênea, incluindo as condições iniciais

4) Resposta à Entrada Nula $y_n(t)$ é semelhante á reposta homogênea, incluindo as condições iniciais

$$y_n(t) = e^{-2t}(A_1\cos(6t) + A_2 sen(6t))$$

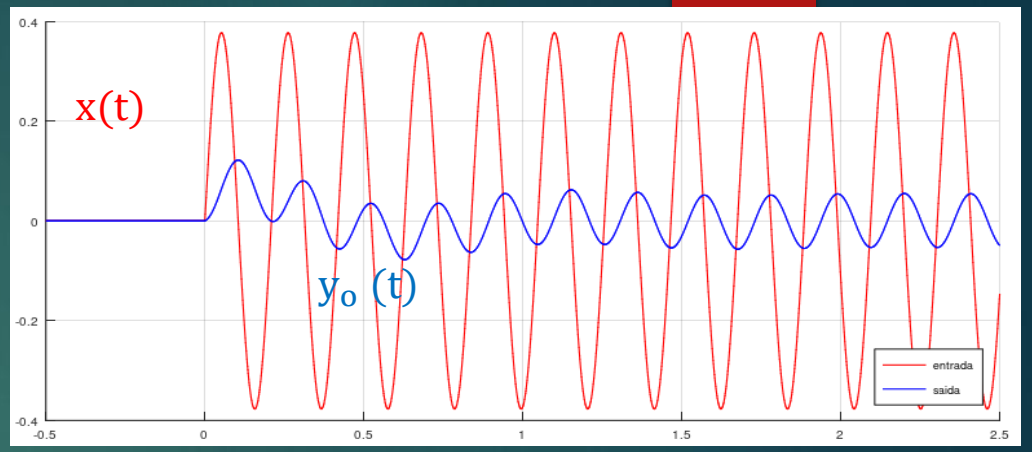
Agora usamos as condições iniciais :

$$y_n(0) = 0.3 = (A_1)$$
$$y_n'(t) = -2e^{-2t}(A_1\cos(6t) + A_2 sen(6t)) + e^{-2t}(-6A_1\text{sen}(6t) + 6A_2 cos(6t))$$
$$y_n'(0) = 0 = -2\,A_1 + 6A_2$$
$$A_1 = 0.3\ ;\quad A_2 = 0.1$$
$$y_n(t) = \mathrm{e}^{-2t}(0.3\cos(6t) + 0.1 sen(6t))$$

```
--> exemplo_C4o()
```

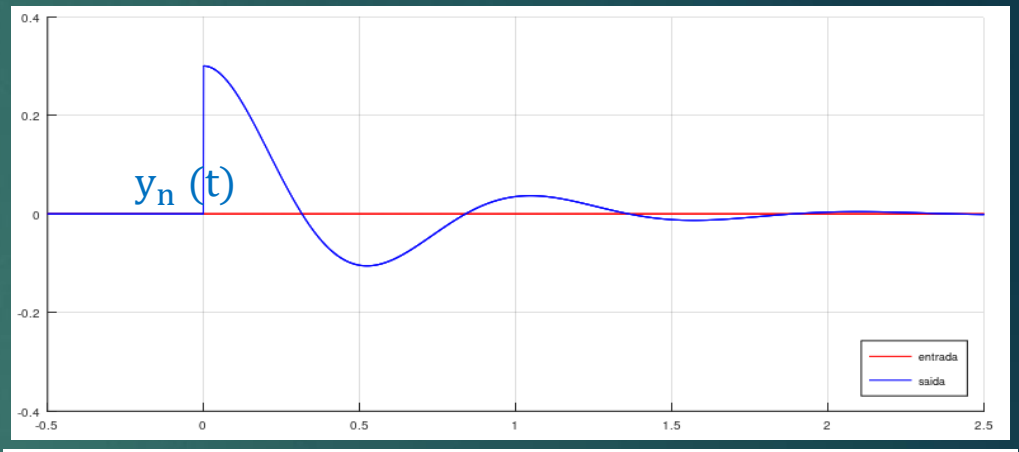
Resposta Homogênea 2ª ordem

1) $\alpha > w \quad y_h(t) = A_1 e^{(-\alpha+\beta)t} + A_2 e^{(-\alpha-\beta)t}$
2) $\alpha < w \quad y_h(t) = e^{-\alpha t}[A_1 \text{sen}(\beta t) + A_2 \cos(\beta t)]$
3) $\alpha = w \quad y_h(t) = e^{-\alpha t}[A_1 + A_2 t]$
4) $\alpha = 0 \quad y_h(t) = A_1 \text{sen}(\beta t) + A_2 \cos(\beta t)$



5) Solução Completa será a soma da Resposta ao Estado Nulo $y_0(t)$ com a resposta à Entrada Nula $y_n(t)$

```
--> exemplo_C4n()
```

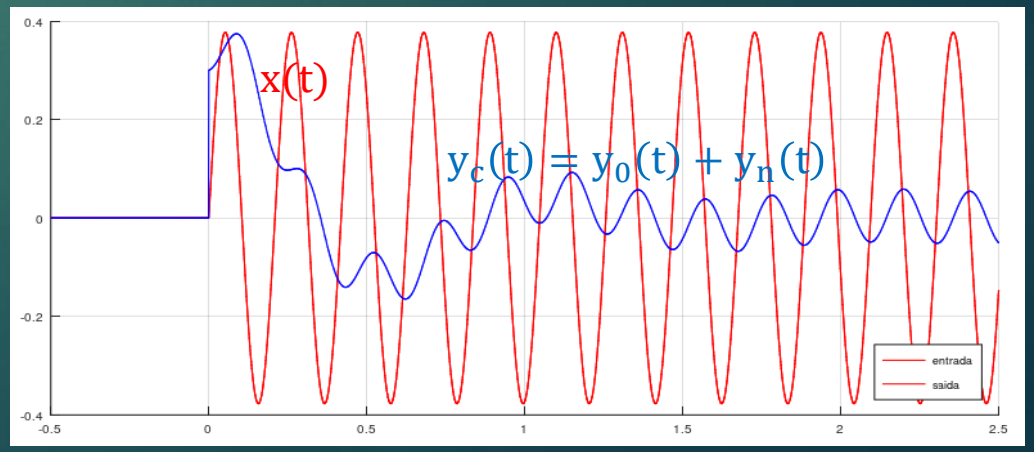


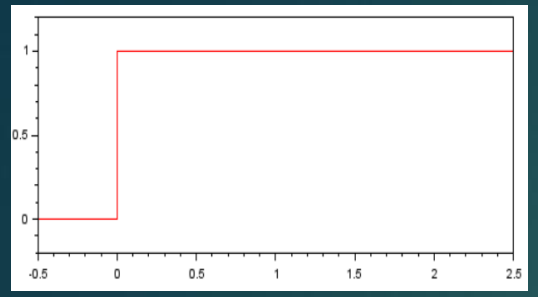
$$y_c(t) = y_{o(t)} + y_n(t)$$

$$y_c(t) = e^{-2t}(0.054\cos(6t) + 0.068\text{sen}(6t)) - 0.054\cos(30t) - 0.001\text{sen}(30t)$$
$$+\mathrm{e}^{-2t}(0.3\cos(6t) + 0.1 sen(6t))$$

$$y_c(t) = \mathrm{e}^{-2t}(0.354\cos(6t) + 0.168\sin(6t)) - 0.054\cos(30t) - 0.001\text{sen}(30t)$$

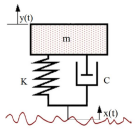
```
--> exemplo_C4c()
```





$x(t) = u(t)$

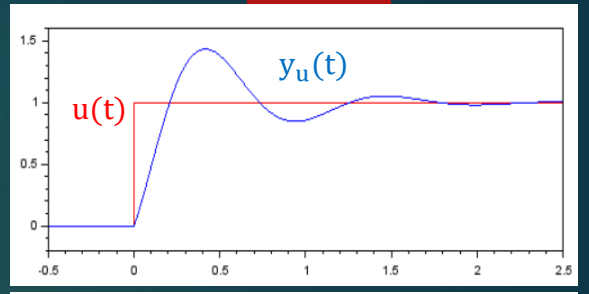
$$y''(t) + 4y'(t) + 40y(t) = 4x'(t) + 40x(t)$$

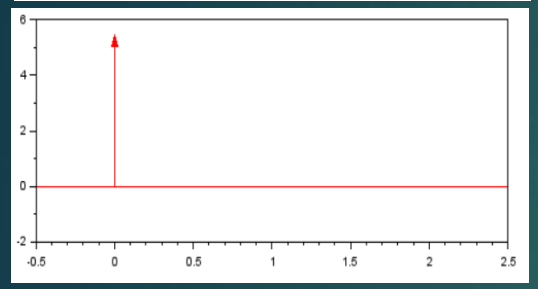


$y(0^-) = 0.0$

$y'(0^-) = 0.0 \;\; -> \; y'(0^-) = 4.0$

$$y = 1 - e^{-2t}\left(\cos(6t) - \frac{1}{3}sen(6t)\right)$$

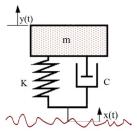




$x(t) = \delta(t)$

$$y''(t) + 4y'(t) + 40y(t) = 4x'(t) + 40x(t)$$

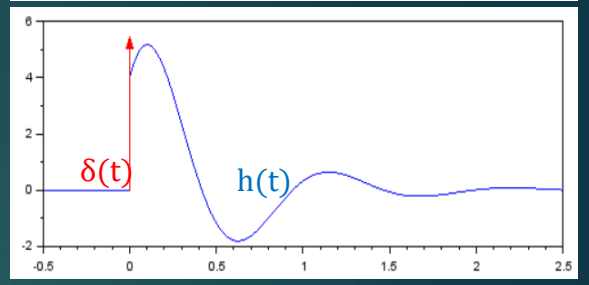
$h(t) = y_u{}'(t)$

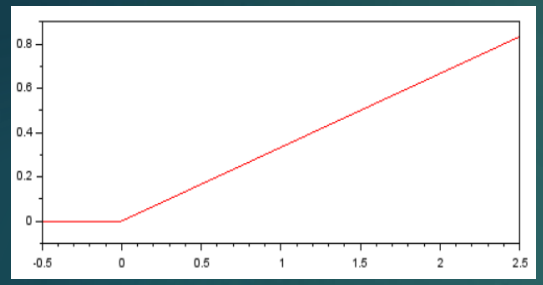


$y(0^-) = 0.0$

$y'(0^-) = 0.0$

$$h(t) = e^{-2t}\left(4\cos(6t) + \frac{16}{3}sin(6t)\right)$$



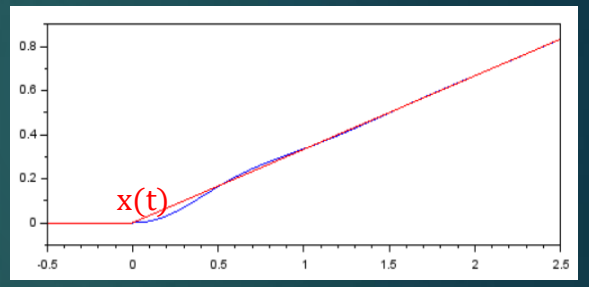


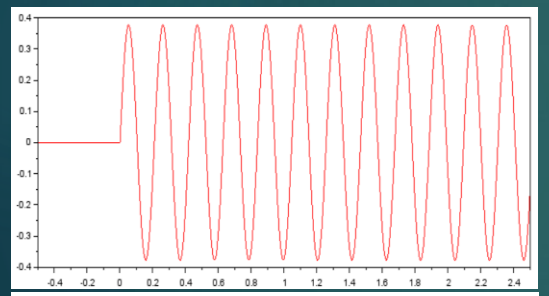
$x(t) = \frac{t}{3}$

$$h(t) = e^{-2t}\left(4\cos(6t) + \frac{16}{3}sin(6t)\right)$$

$$y_0(t) = x(t) * h(t)$$

$$y_o(t) = -\frac{1}{18}e^{-2t}sen(6t) + \frac{1}{3}\,t$$



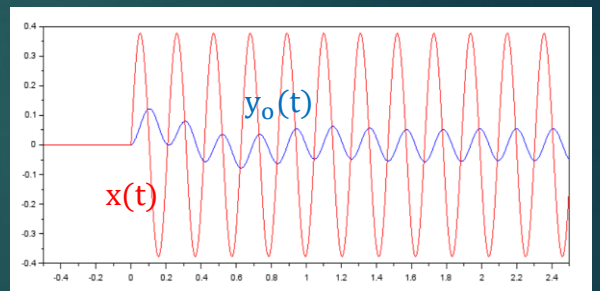


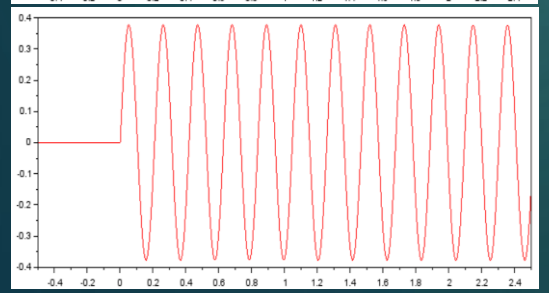
$x(t) = 0.377sen(30t)$

$$h(t) = e^{-2t}\left(4\cos(6t) + \frac{16}{3}sin(6t)\right)$$

$$y_0(t) = x(t) * h(t)$$

$$y_0(t) = e^{-2t}(0.054\cos(6t) + 0.068sen(6t)) - 0.054\cos(30t) - 0.001sen(30t)$$





$x(t) = 0.377sen(30t)$

$$h(t) = e^{-2t}\left(4\cos(6t) + \frac{16}{3}sin(6t)\right)$$

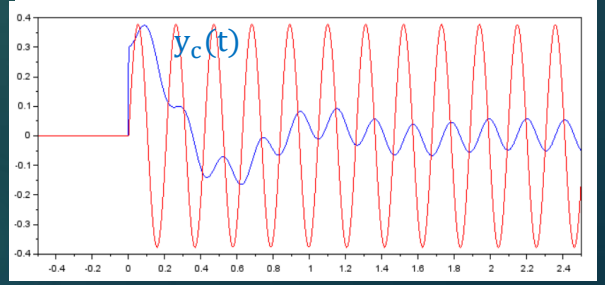
$$y_0(t) = x(t) * h(t)$$

$$y_c(t) = y_{o(t)} + y_n(t)$$

$y(0^-) = 0.3$

$y'(0^-) = 0$

$$y_c(t) = e^{-2t}(0.354\cos(6t) + 0.168\sin(6t)) - 0.054\cos(30t) - 0.001sen(30t)$$



Equação de Diferenças a partir da Equação Diferencial

Transformar um sistema analógico em digital

$$y''(t) + 4y'(t) + 40y(t) = 4x'(t) + 40x(t)$$

$$\boxed{\begin{aligned} &y(t) \cong y[n] \\ &y'(t) \cong \frac{y[n] - y[n-1]}{T} \\ &y''(t) \cong \frac{y[n] - 2y[n-1] + y[n-2]}{T^2} \\ &x(t) \cong x[n] \\ &x'(t) \cong \frac{x[n] - x[n-1]}{T} \end{aligned}}$$

Diferença Finitas

$$\frac{y[n] - 2y[n-1] + y[n-2]}{T^2} + 4\frac{y[n] - y[n-1]}{T} + 40y[n] = 4\frac{x[n] - x[n-1]}{T} + 40x[n]$$

$$y[n] - 2y[n-1] + y[n-2] + 4T\big[y[n] - y[n-1]\big] + 40T^2y[n] = 4\mathrm{T}\big[x[n] - x[n-1]\big] + 40T^2x[n]$$

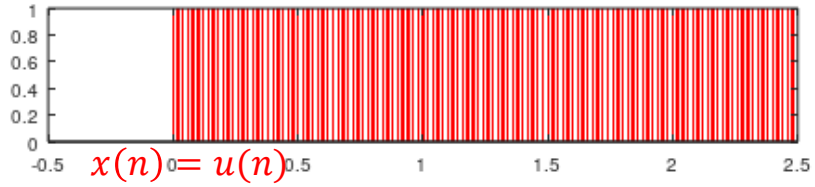
$$(1 + 4T + 40T^2)y[n] + (-2 - 4T)y[n-1] + y[n-2] = (4T + 40\,T^2)x[n] - 4Tx[n-1]$$

$$y[n] = \frac{(2 + 4T)}{(1 + 4T + 40T^2)}y[n-1] - \frac{1}{(1 + 4T + 40T^2)}y[n-2] + \frac{(4T + 40\,T^2)}{(1 + 4T + 40T^2)}x[n] - \frac{4T}{(1 + 4T + 40T^2)}x[n-1]$$

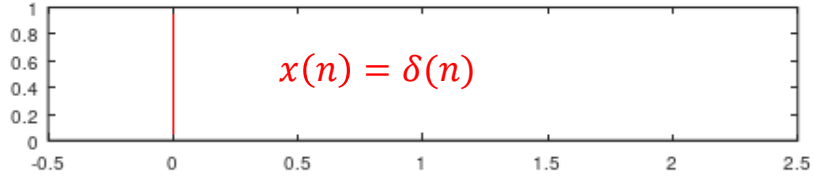
$$T = 0.02$$

$$y[n] = \frac{250}{137}y[n-1] - \frac{125}{137}y[n-2] + \frac{12}{137}x[n] - \frac{10}{137}x[n-1]$$

```
function y_out=Sistema_Mecanico_2aOrdem(x_in,y1,y2,T)
    den=1+4*T+40*T^2;
    a1=(2+4*T)/den;
    a2=-1/den;
    b0=(4*T+40*T^2)/den;
    b1=-4*T/den;
    xi=[0 0 x_in];
    yi=[y1 y2];
    for (i=3:length(xi))
        yi(i)=a1*yi(i-1)+a2*yi(i-2)+b0*xi(i)+b1*xi(i-1);
    end
    y_out=yi(3:length(xi));
endfunction
```

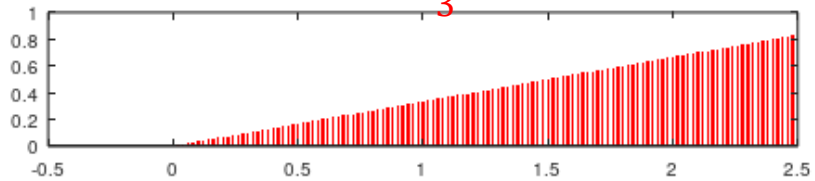


$x(n) = u(n)$

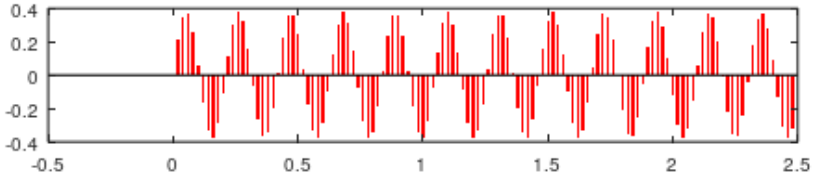


$x(n) = \delta(n)$

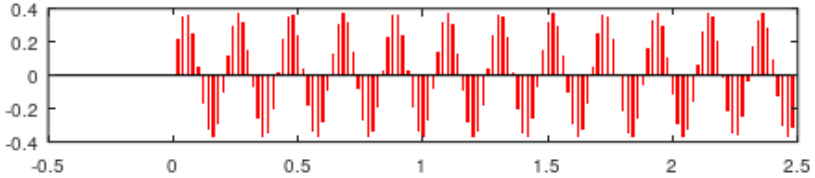
$x(n) = \frac{n}{3}$



$x(n) = 0.377sen(30n)$



$x(n) = 0.377sen(30n)$

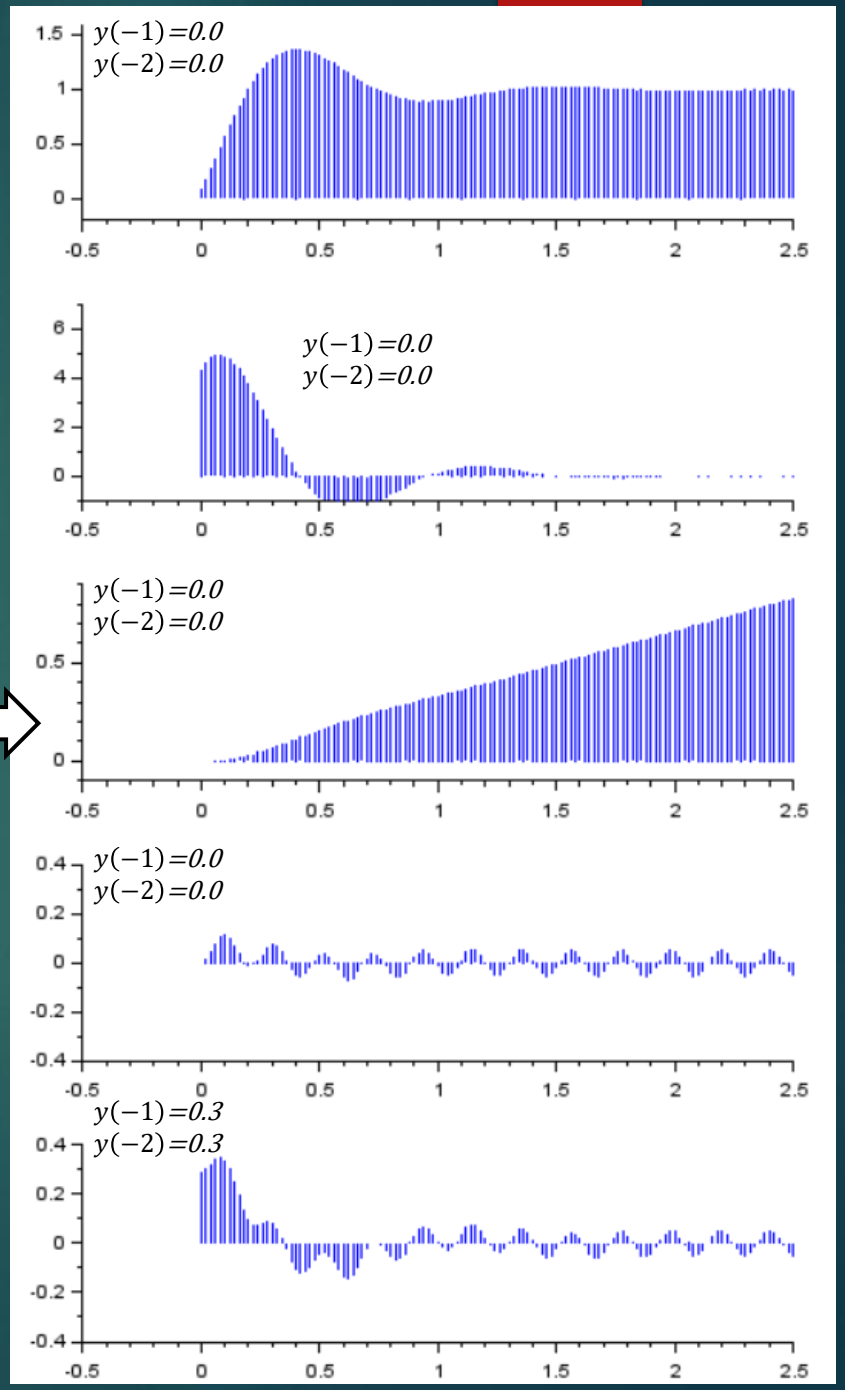


$T = 0.02$

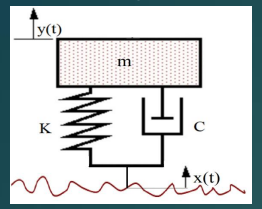
```
--> exemplo_C5(0.02)
```

$$y[n] = \frac{250}{137}y[n-1] - \frac{125}{137}y[n-2] + \frac{12}{137}x[n] - \frac{10}{137}x[n-1]$$

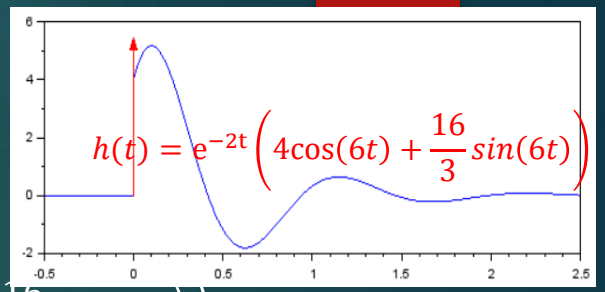
```
function y_out=Sistema_Mecanico_2aOrdem(x_in,y1,y2,T)
    den=1+4*T+40*T^2;
    a1=(2+4*T)/den;
    a2=-1/den;
    b0=(4*T+40*T^2)/den;
    b1=-4*T/den;
    xi=[0 0 x_in];
    yi=[y1 y2];
    for (i=3:length(xi))
        yi(i)=a1*yi(i-1)+a2*yi(i-2)+b0*xi(i)+b1*xi(i-1);
    end
    y_out=yi(3:length(xi));
endfunction
```



a) Analisamos a equação diferencial e encontramos a sua resposta ao Degrau $y_u(t)$ e depois sua Resposta ao Impulso $h(t)$



$$y''(t) + 4y'(t) + 40y(t) = 4x'(t) + 40x(t)$$



b) Encontramos a resposta ao Estado Nulo $y_o(t)$ a uma entrada genérica $x(t)$ por convolução

$$x(t) \Rightarrow \boxed{H(s) = \frac{4s+40}{(s+2)^2+6^2}} \Rightarrow y_o(t)$$

$$y_o(t) = x(t) * h(t) = x(t) * \left(e^{-2t}\left(4\cos(6t) + \frac{16}{3}\sin(6t)\right)\right)$$

c) Caso não seja possível calcular analiticamente a convolução, somos obrigado a encontrar a Resposta ao estado Nulo $y_o(t)$ a uma entrada qualquer $x(t)$, resolvendo diretamente a Eq. Diferencial.

1) $y_o(t) = y_h(t) + y_f(t)$
   - 1.1) $y_h(t)$ de 2ª ordem tem a forma $e^{-3t}(A_1\cos(6t) + A_2 sen(6t))$
     A_1 e A_1 são avaliadas em (1.3)
   - 1.2) $y_f(t)$ – Mesma forma de x(t) com uma constantes B´s
     Encontrar B´s substituindo $y_f(t)$ na eq. diferencial
   - 1.3) $y_o(t) = e^{-3t}(A_1\cos(6t) + A_2 sen(6t)) + y_f(t)$
     Encontrar A_1 e A_1 substituindo as condições nulas em $y_o(t)$

Resposta Homogênea 2ª ordem

1) $\alpha > w$ $\quad y_h(t) = A_1 e^{(-\alpha+\beta)t} + A_2 e^{(-\alpha-\beta)t}$
2) $\alpha < w$ $\quad y_h(t) = e^{-\alpha t}[A_1 sen(\beta t) + A_2 \cos(\beta t)]$
3) $\alpha = w$ $\quad y_h(t) = e^{-\alpha t}[A_1 + A_2 t]$
4) $\alpha = 0$ $\quad y_h(t) = A_1 sen(\beta t) + A_2 \cos(\beta t)$

$x(t) = 1$	$y_f(t) = B$
$x(t) = e^{-kt}$	$y_f(t) = Be^{-kt}$
$x(t) = t$	$y_f(t) = B_1 + B_2 t$
$x(t) = cos(kt)$	$y_f(t) = B_1 \cos(kt) + B_2 \sin(kt)$

d) Como as condições iniciais são nulas, não houve Resposta à Entrada nula $y_n(t)$.

e) Encontramos a saída completa discreta y(n) transformado a equação diferencial em uma equação de diferenças (algoritmo). Para isso tivemos que escolher um período de amostragem T, para transforma a função contínua em discreta.

$$y''(t) + 4y'(t) + 40y(t) = 4x'(t) + 40x(t)$$

$$H(s) = \frac{4s+40}{(s+2)^2+6^2}$$

$$\xrightarrow{T = 0.02}$$

$$y[n] = \frac{1060}{569}y[n-1] - \frac{500}{569}y[n-2] + \frac{69}{569}x[n] - \frac{60}{569}x[n-1]$$

$$H(z) = \frac{69/569 + 60/569z^{-1}}{\frac{500}{569}z^{-2} + \frac{1060}{569}z^{-1} + 1}$$

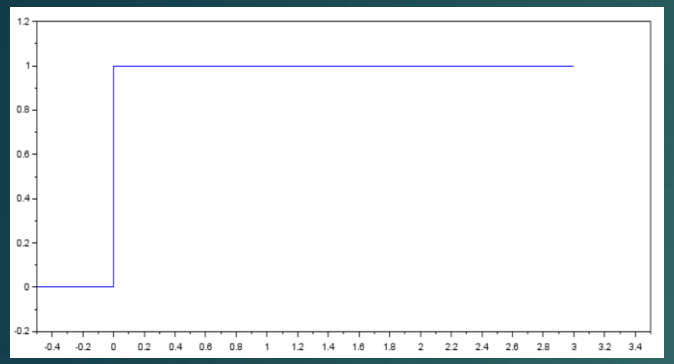
Trocar o amortecedor do Sistema

$$x(t) = u(t)$$

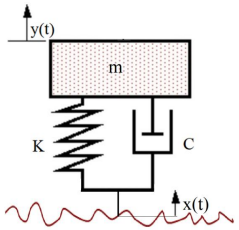
$$y''(t) + \frac{C}{m}y'(t) + \frac{K}{m}y(t) = \frac{C}{m}x'(t) + \frac{K}{m}$$

$$y''(t) + 4y'(t) + 40y(t) = 4x'(t) + 40x(t)$$

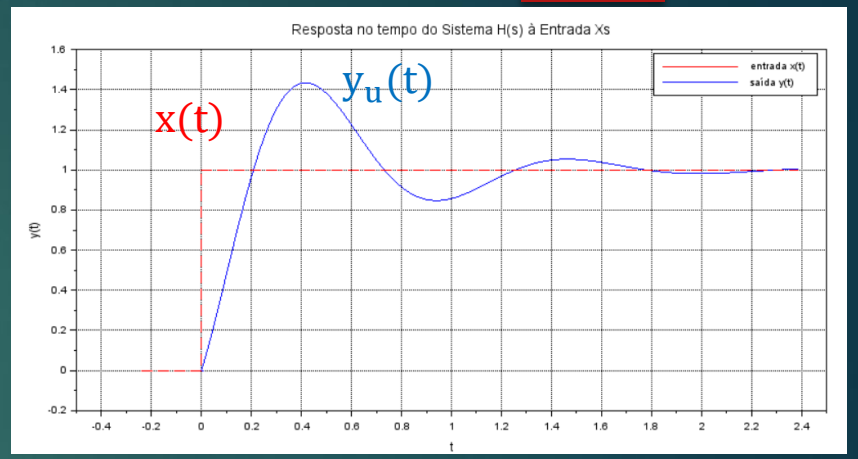
$$y_u(t) = 1 - e^{-2t}\left(\cos(6t) - 0.33sen(6t)\right)$$



m=300kg
K= 12000 N/m
C=1200 N/(m/s)



$$H(s) = \frac{4s + 40}{(s+2)^2 + 6^2}$$



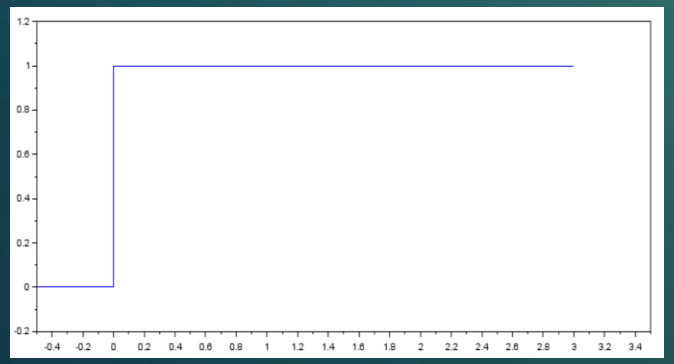
- *Sugere-se aumentar o amortecer para C=2400N/(m/s)*

$$y''(t) + \frac{C}{m}y'(t) + \frac{K}{m}y(t) = \frac{C}{m}x'(t) + \frac{K}{m}$$

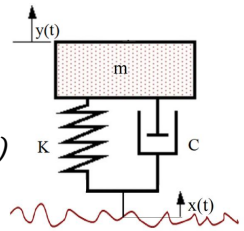
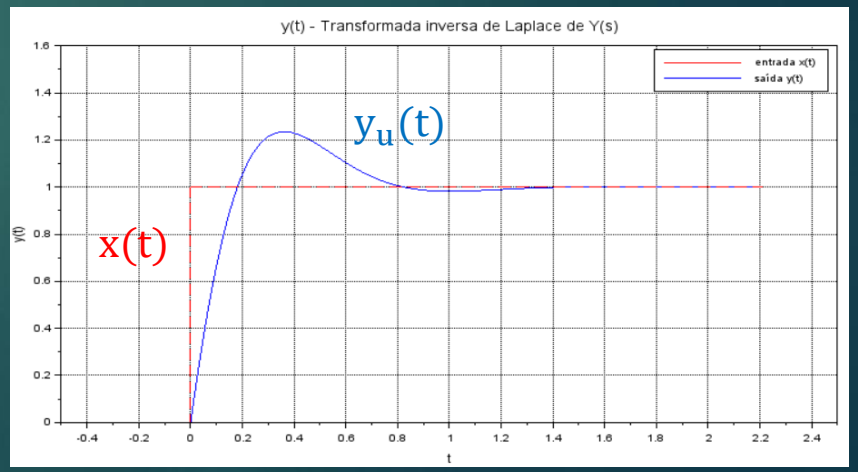
$$y_u(t) = 1 - e^{-4t}\left(\cos(4.9t) - 0.82sen(4.9t)\right)$$

$$x(t) = u(t)$$

$$y''(t) + 8y'(t) + 40y(t) = 8x'(t) + 40\,x(t)$$



m=300kg
K= 12000 N/m
C= 2400N/(m/s)



$$H(s) = \frac{8s + 40}{(s+4)^2 + 4.9^2}$$